

UNIVERSITÀ DEGLI STUDI DI ROMA

“TOR VERGATA”



TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

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ENGINEERING SCIENCES

THESIS IN
FEEDBACK CONTROL SYSTEMS

**“REFERENCE GENERATOR FUNCTION IDENTIFICATION FOR ELECTRICAL
VEHICLES WITH IN-WHEEL MOTORS”**

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ABSTRACT

In recent times, motor speed reference generators have been developed to enhance the cruise control of electric vehicles. These generators are designed for both centralized electric motors, which are suitable for straight maneuvers, and in-wheel motors, which are ideal for bend maneuvers with small constant steering angles. The main aim of these generators is to achieve steady-state operation at a safe tire longitudinal slip without prior knowledge of any specific external conditions.

An innovative solution has been proposed in recent works, which utilizes Pacejka-like curves to represent the torque current-slip characteristics of the vehicle. The solution involves designing a new contraction-mapping-based automatic tuning procedure for the longitudinal velocity of the vehicle. This approach is a significant improvement over the previous conservative approach that produced unduly small longitudinal velocities and yaw rates under relatively favorable external conditions.

The new approach guarantees a safe operating condition close to the maximum of the torque-slip characteristics, without any prior knowledge of the tire-road adhesion coefficient. Comparative simulations using CarSim illustrate the effectiveness of this approach, even in the presence of uncertain parameters and complex vehicle dynamics that are typically neglected during the control design stage.

In summary, the proposed approach offers a more effective and innovative solution to the problem of motor speed reference generators for the cruise control of electric vehicles. It provides a safe and efficient means of achieving steady-state operation, even in the absence of prior knowledge of specific external conditions, and it can account for uncertain parameters and complex vehicle dynamics that may be overlooked during the control design process.

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1) Introduction

1.1) Motive of the Thesis

According to statistics from the UN, 1.25 million people worldwide pass away and up to 50 million are injured in traffic accidents each year. Around 95% of all traffic accidents in the EU are caused by human error, and 25,300 individuals died on Union roads in 2017 alone. Children, the elderly, and people with physical disabilities have little to no access to personal mobility every year. Additionally, by doing away with the never-ending search for a parking place, which studies suggest can account for up to 30% of inner-city traffic, half a million metric tons of CO₂ emissions might be avoided annually in Germany alone. In addition to significantly lowering these numbers and enhancing road safety, driverless cars can also significantly lower traffic congestion and greenhouse gas and air pollution emissions. Increasing accessibility to road transportation for the elderly and people with impairments or decreased mobility is one way to increase mobility. Due of this, a lot of academic studies and automakers have been concentrating recently on the creation of ADAS (Advanced Driver Assistance Systems) and self-driving cars.

ADAS is defined as "vehicle-based intelligent safety systems which potentially improve road safety in terms of crash avoidance, crash severity mitigation and protection, and post-crash phases" by the European Road Safety Observatory (ERSO). Integrated in-vehicle or infrastructure-based systems that contribute to more than one of these crash stages might be referred to as ADAS. Autonomous driving vehicles can be thought of as those that carry out transportation tasks without the need for human intervention, using algorithms carried out by an on-board computer to mimic the actions and decision-making of the driver. Environment perception, planning and decision-making, and vehicle control are the three major categories into which the key competencies of an autonomous vehicle software system can be divided. By using external sensors like lasers, cameras, and radar, the environment perception module gathers information about its surrounds. It then combines the data by creating environment maps to identify drivable surfaces. The targeted course and the speed profile are determined by the planning and decision-making module, which also collects, manages, and combines

task information with data on vehicle states and drivable surfaces. By sending the signal required for actuators to lead the autonomous vehicle, the vehicle control module coordinates the engine, brakes, and steering to track the desired course and speed.

Autonomous driving vehicles can be thought of as those that carry out transportation tasks without the need for human intervention, using algorithms carried out by an on-board computer to mimic the actions and decision-making of the driver. Environment perception, planning, and decision-making are the three main categories into which the key competencies of an autonomous vehicle software system can be divided. and vehicle management. By using external sensors like lasers, cameras, and radar, the environment perception module gathers information about its surrounds. It then combines the data by creating environment maps to identify drivable surfaces. The targeted course and the speed profile are determined by the planning and decision-making module, which also collects, manages, and combines task information with data on vehicle states and drivable surfaces.

1.2) Classification of autonomous vehicles

There are six levels of autonomous driving, each with a different level of automation. The Society of Automotive Engineers (SAE) has classified the spectrum as ranging from level 0 to level— from no automation to fully autonomous vehicles.

At a high level, you can think of the levels of automation as:

- Level 0: No automation at all, the vehicle may temporarily assist with steering or speed control, but the human driver is in charge of all aspects of driving.
- Level 1: Very light automation (cruise control, etc.), Because the vehicle takes over either the speed or the steering control on a sustained basis, longitudinal or lateral support is guaranteed in well-defined driving scenarios.

- Level 2: Some automation but always requires human attention, The vehicle assumes control of both speed and steering, ensuring continuous longitudinal and lateral support under well-defined driving scenarios.
- Level 3: Can self-drive but require intervention in severe conditions, The vehicle becomes capable of taking full control in well-defined driving scenarios, but the driver must always be prepared to regain control when the system requires it.
- Level 4: Highly autonomous, Human interaction is no longer required; the vehicle takes full control and completes the journey autonomously under limited driving scenarios. Pedals and a steering wheel are likely to remain to ensure the ability to drive in scenarios that go beyond the defined use.
- Level 5: Completely autonomous, under all driving scenarios, the vehicle assumes complete control; no provisions for human control remain. The journey concept will be disruptively innovated, and the entire vehicle design will be revolutionized.

1.3) A Possible Application of Autonomous Vehicles

The prospective uses of autonomous vehicle technology, such as automated roads and urban transit, have recently attracted the attention of the automobile industry. Fully automated driving is still a difficult endeavor that calls for expertise in areas like modeling, automatic control, trajectory development, and image processing. It also contains hard features. For vehicle guidance, including steering and velocity control, the latter issue is of utmost significance. The steering and velocity tracking problems are discussed independently or in conjunction in the literature, as will be demonstrated as being state of the art. Electric automobiles, buses, trains, and subways are at the cutting edge of transportation technology that is secure, environmentally friendly, and energy efficient. Electric motors are used to power them because of their outstanding serviceability, durability, high efficiency, high power density, high torque to inertia ratio, low noise, low pollution, and quick torque response. Several different layouts are conceivable inside this framework: two axial motors, one central motor, and one motor each axle (as in the Jaguar I-PACE and Tesla model S). Three motors are used in vehicles like the Audi E-Tron, which has two decoupled coaxial motors on the rear axle that provide differential

torque and one front motor for propulsion. In-wheel motors result in the following: straightforward packaging of the drivetrain; bulk reduction and improved energy efficiency; and substitution of the electric differential for the mechanical differential. Yet, the entire research community has typically become interested in certain control and estimation problems. It is suggested that the reader go to for a thorough discussion of them.

The regulation of the vehicle longitudinal velocity v_x to a specified constant vehicle velocity reference v_x is a particular issue with autonomous vehicles and is commonly referred to as cruise control. Regarding the aforementioned cruise control problem and its variant, known as "adaptive cruise control," in which a safe distance must also be maintained between the vehicle and the one in front of it, numerous contributions have been made, as noted in. Several literary sources include related material.

1.4) A determined issue

Now, the issue of controlling the motor speed to a properly defined motor speed reference can be used to solve the cruise control problem for a vehicle with a centralized electric motor (and fixed transmission gear ratio) and make straight moves. The sole condition is that the work of regulating the electric motor's speed is swiftly accomplished. This is the direction of the solution has then been expanded to the slip constraint case in, where the vehicle longitudinal velocity reference is automatically (internally) adjusted via feedback in response to uncertain external conditions like tire-road adhesion coefficients and road slopes. In this case, the vehicle longitudinal velocity reference is not just exogenously given, but it is also adjusted in response to the slip constraints. With no prior knowledge of the occurrence of a specific external condition, the contraction theorem has been specifically applied to a smooth desired design function in the (ω_L, v_x) -plane to drive the pair (ω_L, v_x) into safe tire longitudinal slip-regions: the vehicle velocity is automatically reduced when an abrupt change in tire-road adhesion coefficient uncertainly occurs. In addition, whose contribution expands on the strategy predicted in, has further developed such a strategy to address the more general

case of electric rear-wheel drive vehicles that are outfitted with two independent in-wheel rear motors (one per tire) and are required to perform a bend maneuver (with sufficiently small constant steering angle). However, the contraction mapping algorithm is used to decrease the speed in correct way.

Once the rotor speeds are seen as independent control inputs, the two motors can function as a rear differential while also providing propulsion. Then, all that is needed to naturally have ω_L perform the role of in the single motor arrangement is to use a kinematic constraint to make the differential component of the wheel speeds, ω_D , proportionate to the longitudinal component, ω_L . All of the aforementioned procedures, however, are structurally conservative because the smooth desired design function, which governs the convergence of the pair (ω, v_x) - (ω_L, v_x) , respectively, into safe tire longitudinal slip-regions regardless of the external condition, by design ensures excessively small values of the longitudinal velocity (and yaw rate) under relatively favorable external conditions. To overcome the aforementioned structural disadvantage, an innovative solution is presented. It makes clever use of Pacejka curves to create a new procedure that automatically modifies longitudinal velocity (and yaw rate). This procedure is inspired by the recent definition of wheel angle while taking advantage of the converging load torque estimates provided by the motor speed controls. Such a wheel angle employs the simultaneous force/slip information provided by a coordinate pair on the tire force-slip characteristics curve: the angular deviation of a straight line passing through the origin of the force/slip plane and a point on such a plane representing a steady-state operation.

MATLAB and Simulink have been used to implement the overall autonomous driving system. The method uses a model stereo camera that, as explained in next chapter, uses synthetic data from a model driving scenario for lane detection. In the actual implementation, this data is derived by a lane detection algorithm based on the streaming of stereo camera data in real-time. The lane boundaries information is taken from three simulated driving situations that replicate a highway, inter-urban road, and urban road because in our instance the simulations are carried out in MATLAB/Simulink. A rigid, three-degree-of-freedom vehicle model is used to simulate the vehicle's dynamics.

1.5) CarSim

CarSim is a software application that simulates the dynamic behavior of passenger cars and light-duty trucks. It employs 3D multibody dynamics models to accurately reproduce the physics of the vehicle in response to driver and/or automation controls such as steering, throttle, braking, and gear shifting. A 3D ground/road surface, as well as aerodynamic and wind effects, are examples of environmental conditions.

Simulink S-Functions are included in CarSim to connect the CarSim solver library to Simulink models. The diagram below depicts the S-Function vs sf in a Simulink model. The block is linked to the rest of the model via Import and Export variables that were enabled in CarSim to meet the Simulink model's requirements. When the Simulink model is run, the S-Function loads the CarSim VS Solver library, constructs a VS Math Model using VS Modularity functions, and exchanges I/O data with Simulink at each time-step. Simulink plots can display the results as the simulation runs. Following the completion of the simulation, VS Visualizer can be used to further analyze data and view the animated video.

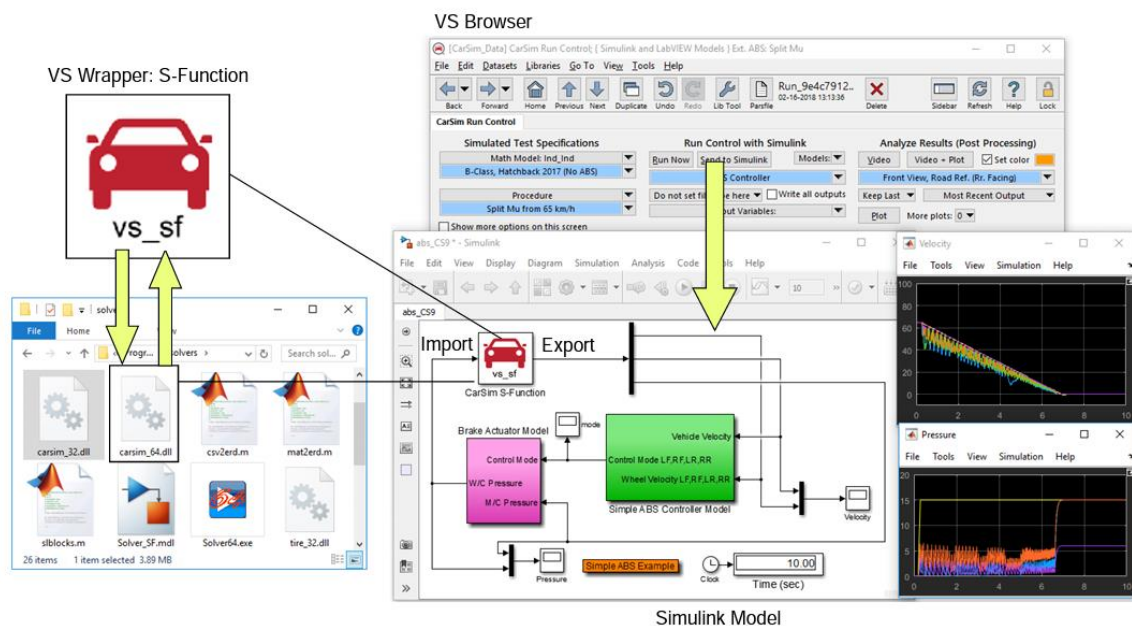


Fig.1.5.1 CarSim and Simulink

Vehicle Models with High-Fidelity Systems: CarSim math models are represented at the system level, which means that the vehicle data is intended to be measured or calculated and does not require detailed knowledge of component materials, suspension linkage intricacies, and so on. The vehicle's mathematics are detailed enough that the simulation can replicate physically measured responses within the limits of testing repeatability.

Parametric Vehicle Definition: CarSim represents the vehicle using a combination of parameters and variables. Parameters stand in for quantifiable qualities like size and inertia properties, as illustrated below. Configurable Functions use a range of extrapolation and interpolation techniques to tie model variables to linear coefficients or tables. Many of the tables provide data from tire and suspension test rigs. Other potentially nonlinear relationships that can be imported from spreadsheets, including road geometry and friction, are likewise conveniently represented in tabular form. Figure.1.5.2

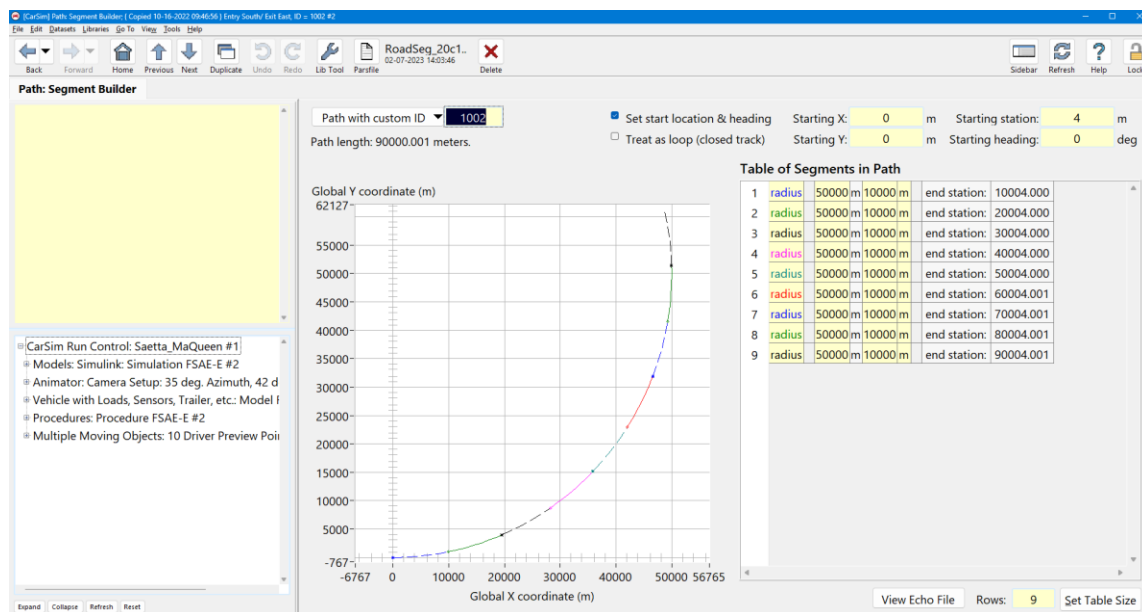


Fig.1.5.2 Parametric and tabular data used in CarSim

Quick Calculations: A simulation of a 60-second test will conclude in 3 or 4 seconds on a recent Windows machine since the CarSim math model is highly optimized for calculation performance and often works 15 to 20 times faster than Windows clock

time. Under conditions of heavy load, such as the addition of numerous moving objects and sensors and the need to save thousands of outputs to files, the simulation's execution speed may be slowed down. The simulations on Windows nonetheless operate faster than real time.

Plots and Visualization: CarSim includes VS Visualizer, a tool for creating high-quality animations of simulation results corresponding to engineering plots of hundreds (or thousands) of variables from the math model. The plots and video are time-synchronized to make viewing qualitative and quantitative data easier, and the ability to overlay numerous runs (up to six in total) enables comparisons between various vehicle and control combinations. Figure.1.5.3

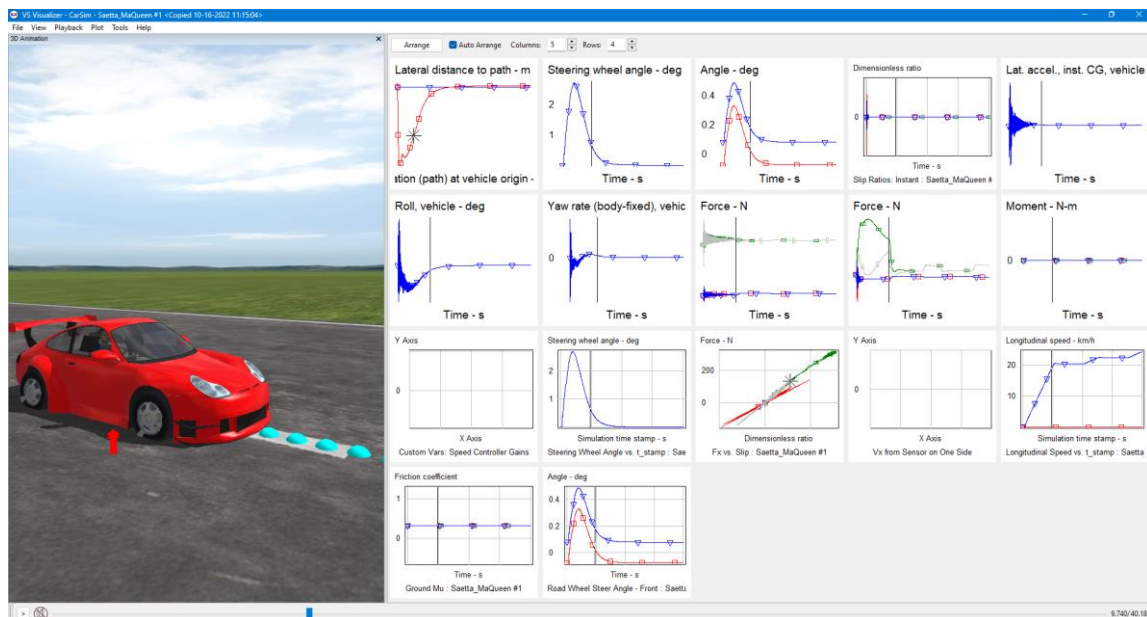


Fig.1.5.3 Viewing synchronized video and plot of CarSim vehicle response with VS Visualizer

ADAS and AV Support: CarSim includes moving "target" objects that represent traffic vehicles, pedestrians, bicycles, and other objects in order to simulate scenarios involving Advanced Driving Assistance Systems (ADAS) and/or autonomous vehicles (AVs). These target objects are detected by virtual sensors, with each sensor providing 24 calculated output variables for each possible sensor / target detection pair.

Database, User Interface, and Documentation: CarSim includes over 600 example simulations, each of which includes one or more vehicles and a set of test conditions. These examples include over 40 vehicles representing at least ten distinct vehicle configurations.

1.6) The Fundamentals of CarSim

If you are running with the Browser and VS Visualizer on Windows, there are several fundamental functionalities in CarSim that may be used for nearly any application. You may view the data that was used to create the simulations as well as the results of previous simulations using CarSim. Moreover, you can construct new instances by copying and editing old ones to create new simulations utilizing existing car examples.

On CarSim's home page you can navigate to libraries. Libraries in the database folder contain the CarSim GUI displays. The title of the dataset being displayed is displayed in the Windows title bar along with the name of the current library.

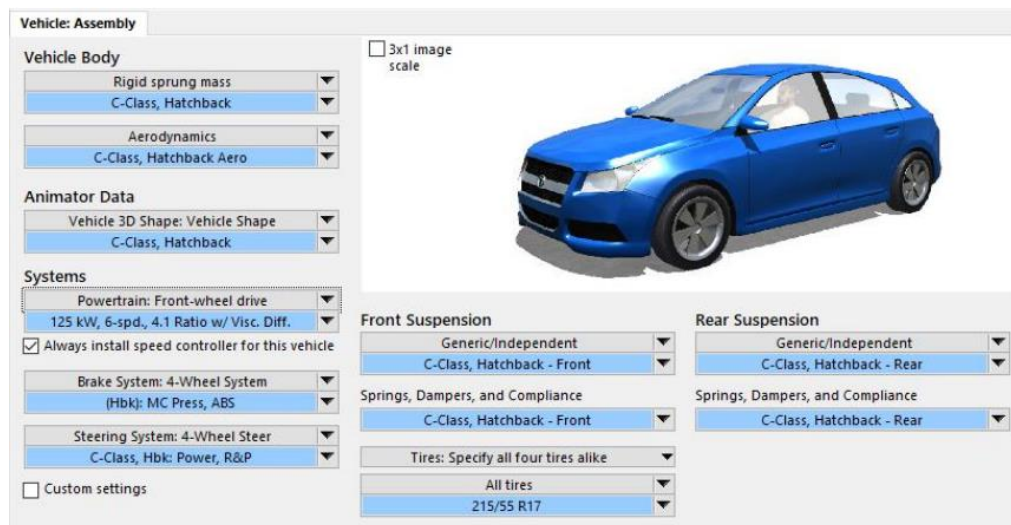


Fig.1.6.1 A *Vehicle: Assembly* dataset that was used in the example simulation setup.

Below the navigation bar, a row of buttons can be found. The existing dataset can be copied using the Duplicate button before being modified to make room for additional data. If the dataset is enabled, locations with editable text are indicated with a yellow backdrop. The section on the left that features optional notes about

the current dataset and a tree view of the current simulation is shown or hidden using the Sidebar button.

The GUI makes extensive use of blue configurable hyperlinks, which function similarly to Web browser hyperlinks. Hovering the mouse cursor over a blue link will underline the text in the same way that a web link does.

1.7) Co-Simulation and Third-Party Software

All of the features described thus far are included in any CarSim installation, with the exception of CarSim for non-RT Linux, which does not include a VS Browser. CarSim users frequently use it in conjunction with other (third-party) software. CarSim's architecture is designed to facilitate the creation of stacks using CarSim and other tools such as MATLAB-Simulink, Unreal Engine, Python, and others. *From Help menu item: Release Notes > System Requirements.* The System Requirements and Compatibility documentation explains what other software can be used with CarSim.

Import and Export Arrays: Running a CarSim math model from another program is as simple as loading the VS Solver library and pressing the Run button. However, the CarSim model is usually part of a larger simulation and must communicate information to and from the other software. This is accomplished through the use of arrays of Import and Export variables. Existing variables are used to configure the Import and Export arrays. A typical setup for running under external software will activate a subset of the available variables for import and export, matching the external model's requirements. If the external model, for example, has a braking intervention system, it should provide throttle and brake controls as well as sensor outputs from CarSim.

Simulink and LabVIEW: CarSim includes Simulink S-Functions to connect the CarSim solver library to Simulink models. The following diagram depicts the S-Function vs sf in a Simulink model. The Import and Export variables, which were enabled in CarSim to meet the Simulink model's requirements, connect the block to the rest of the model. When the Simulink model is executed, the S-Function loads the CarSim VS Solver library, builds a VS Math Model with VS Modularity functions, and exchanges I/O data with Simulink at each time-step. Simulink plots can display the

results of simulations as they run. After the simulation is finished, VS Visualizer can be used to further analyze data and view the animated video.

2) Modelling and Regulation

In reference to the entire model, this chapter covers the mathematical modeling of the vehicle dynamics for validation and control synthesis. The quality of the models offered has a big impact on model-based control. On the one hand, accurate models often require expensive computations and produce precise forecasts. Simple models, on the other hand, require less processing but yield predictions that are less precise. As simulations will be used to assess the identification of the reference generator in electrical cars with in-wheel motors, a validation model is required. The validation model must accurately reflect the characteristics of an actual vehicle.

In the first part of this section, the finding in one of the research is theoretically generalized to electric rear-wheel drive vehicles that use two in-wheel motors and must perform bend maneuvers (with constant curvature radius). With the definition of appropriate motor speed references, the issue is to control both the longitudinal speed and the yaw rate to supplied constant references. The yaw rate and longitudinal speed of the vehicle are presumed to be delivered by a CAN-bus in this scenario (Controller Area Network).

Electric rear-wheel drive vehicles that require bend maneuvers and employ two in-wheel motors are taken into account. The following is the terminology for the vehicle parameters: R : the wheel's radius; l_w : the car's width; l_f : the center of mass's distance from the front axle; l_r : the center of mass's distance from the rear axle; and δ : the steering angle. The focus of the research is on vehicles with rear wheel drives, also known as rear-left (r, l) and rear-right (r, r) drives. The vehicle's steering dynamics and suspensions-dependent dynamics are, in first approximation, ignored, and no-load transfer is taken into account on each wheel. The vehicle is assumed to drive across a relatively flat plane.

Simplifying the following assumptions:

- a rear-wheel drive vehicle is taken into consideration, with front steering wheels included.
- the vehicle moves on a relatively flat plane; the vehicle steering dynamics and the suspensions-dependent dynamics are, in first approximation, neglected.
- no load transfer is considered on each wheel.
- the in-wheel motors are ideal actuators, allowing the angular speeds ω_{r1} and ω_{r2} to be calculated.

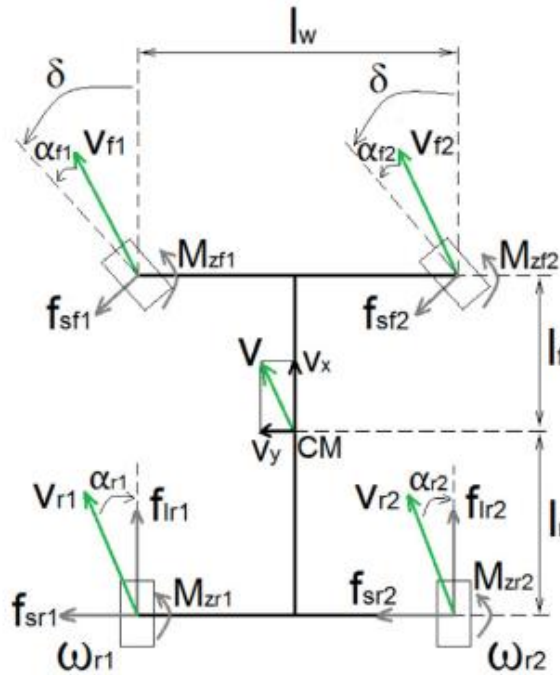


Fig.2.1 Full vehicle model

These presumptions enable an approximate analysis of the vehicle motion while maintaining a certain level of design simplicity. These will be eliminated in simulations (and experiments), where intricate dynamics and useful considerations will be taken into account. For instance, the angular speeds ω_{r1} and ω_{r2} , which are being used as control inputs, will develop into sufficiently smooth rotor speed references in simulations and experiments allowing them to be incorporated into the speed tracking controls powering each in-wheel electric motor.

2.1) Reference for Provided Longitudinal Velocity

Think about the challenge of defining appropriate motor speed references and limiting the longitudinal velocity v_x and yaw rate r to constant references v_{x*} and r .

Let the associated regulation error be denoted by $v_x = v_x - v_{x*}$, $\tilde{r} = r - r_*$ (typically available for feedback through the CAN-bus).

$$\omega_L = \frac{\omega_{r,l} + \omega_{r,r}}{2}$$

$$\omega_D = \omega_{r,r} - \omega_{r,l}$$

The longitudinal and differential components of the wheel speeds, which are made individually responsible for the regulation of v_x and r , respectively, are denoted using an invertible transformation. With certain technical constraints, relatively tiny tire longitudinal slippage $\lambda(r, l)$ and $\lambda(r, r)$

$$\lambda_i = \left(\frac{R_{\omega i} * \omega_i}{v_x} - v_x \right)$$

$R_{\omega i}$ is the radius of the wheel, $i \in \{(r, l), (r, r)\}$,

the proportional-integral decoupled controller on the motor speeds, with

i) a suitably high curvature radius of the bend maneuver, and

ii) the wheel speeds (the wheel centers are rigidly connected to the vehicle chassis)

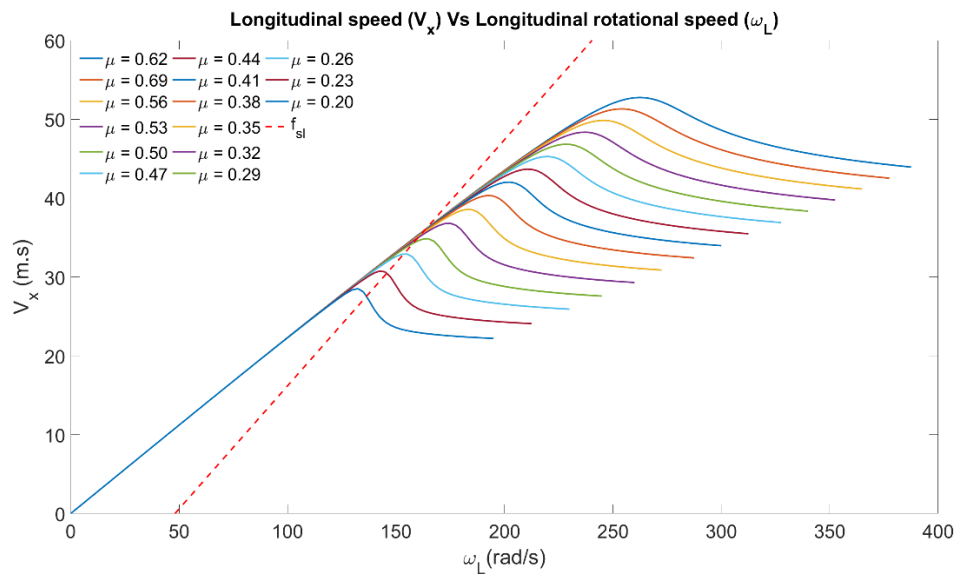


Fig.2.1.1 Longitudinal speed vs Longitudinal rotational speed

The existence of constant reference inputs ω_L^* and ω_D^* and of a constant reference lateral velocity v_{y^*} , being compatible with the constant output references v_{x^*} and r^* , is further assumed. Such constant references ω_L^* , ω_D^* , v_{y^*} are uncertain owing to the overall system uncertainties. Let (again) $\tilde{v}_x = v_x - v_{x^*}$ and $\tilde{r} = r - r^*$ be the longitudinal speed and yaw rate regulation errors. The proportional–integral control

$$\begin{aligned}\omega_L &= \frac{v_{x^*}}{R_\omega} - k_{px} \tilde{v}_x - k_{ix} \int_0^\cdot \tilde{v}_x(\tau) d\tau \\ \omega_D &= \frac{l_w r^*}{R_\omega} - k_{pr} \tilde{r} - k_{ir} \int_0^\cdot \tilde{r}(\tau) d\tau\end{aligned}$$

Here k_{px} , k_{ix} , k_{pr} , k_{ir} are constant positive proportional–integral gains, Whereas the constants v_{x^*} / R_ω and $l_w r^* / R_\omega$ represent (ideal) kinematically approximating values for the uncertain ω_L^* and ω_D^* , respectively. v_{x^*} , given constant reference for the longitudinal velocity and r^* , constant reference for the yaw rate r .

The control inputs ω_L and ω_D are thus separately responsible for the regulation of v_x and r , respectively. Furthermore, ω_L and ω_D univocally define the control inputs ω_{r1} and ω_{r2} , owing to the fact that the transformation is invertible. The constant reference for the yaw rate r .

$$r_* = \left(\frac{\delta}{l_f + l_r} v_{x^*} \right)$$

The above equation here is currently presented. It is described in terms of the longitudinal velocity v_x 's constant reference, or v_{x^*} .

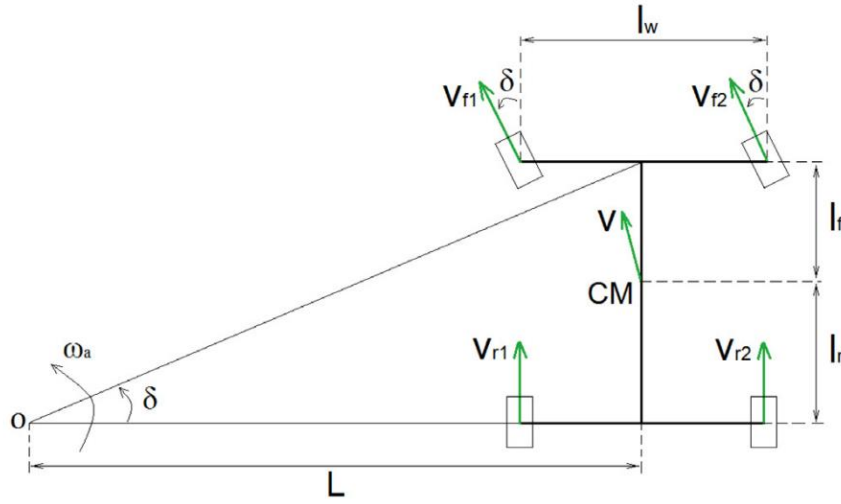


Fig.2.1.2 Ideal kinematic steering with wheels in free rolling

It is founded on kinematic justifications. From, Figure.2.1.2 as the wheelbase is considerably smaller than the curvature *radius* R_c of the vehicle's center of mass (CM) with respect to the center of the curve (O), the following approximations are defined for modest steering angles:

$$R_c = (l_f + l_r) \cos(\delta) = \left(\frac{l_f + l_r}{\delta} \right)$$

2.2) Automatic Decreasing of Longitudinal Velocity

The previous subsection's vehicle velocity reference v_x is predetermined. It is not automatically adjusted via feedback in response to an uncertain tire-road adhesion coefficient, so tire longitudinal slips at steady-state may not belong to safe regions. Slip constraints are considered in the second part of, and a secondary procedure, generalizing the one-off, is proposed to be combined with the one mentioned in the previous subsection. The main idea is to limit the design to the (ω_L, v_x) -plane once the kinematic ω_L -constraint on ω_D :

$$\omega_D = \frac{l_w r_*}{R_\omega} - k_{pr} \tilde{r} - k_{ir} \int_0 \tilde{r}(\tau) d\tau$$

$$\omega_D \equiv \frac{l_w r_*}{R_\omega} = \frac{l_w v_{x*} \delta}{R_\omega (l_f + l_r)} = \frac{l_w \omega_L \delta}{(l_f + l_r)}$$

Once ω_L is replaced, the scenario of is reproduced: for given external conditions (denoted by), the (ω_L, v_x) -steady-state operating conditions define a specific (ω_D -constrained) steady-state curve $g_{\mu,c}(\cdot)$ in the (ω_L, v_x) -plane, and a contraction theorem-based approach similar to the one qualitatively presented in guarantees convergence to the intersection points $P_{\mu,c}$ (characterized by safe tire longitudinal slip values) between $g_{\mu,c}(\cdot)$ and a suitable smooth, strictly increasing function $f_{sl}(\cdot)$ where s_l stands for safe slip.

2.3) Vehicle Model

A dynamical nonlinear model of an electric vehicle with two in-wheel motors is given, with the following motion equations.

$$\begin{aligned} \dot{v}_x &= r v_y + \frac{1}{m} [-(f_{sf1} + f_{sf2}) \sin(\delta) - c_a v_x^2] \\ &\quad + \frac{1}{m} (2a(v_x, v_y) \omega_L - b) \\ \dot{v}_y &= -r v_x + \frac{1}{m} [(f_{sf1} + f_{sf2}) \cos(\delta) + f_{sr1} + f_{sr2}] \\ \dot{r} &= \frac{l_f}{J_z} (f_{sf1} + f_{sf2}) \cos(\delta) - \frac{l_r}{J_z} (f_{sr1} + f_{sr2}) \\ &\quad + \frac{l_w}{2J_z} (f_{sf1} - f_{sf2}) \sin(\delta) + \frac{1}{J_z} \sum_{j \in \{f1, f2, r1, r2\}} M_{zj} \\ &\quad + \frac{l_w}{2J_z} a(v_x, v_y) \omega_D, \end{aligned}$$

In a vehicle dynamics model, the longitudinal and lateral speeds, along with the yaw rate, are crucial parameters to describe the motion of a vehicle. Additionally, the angular speed of each wheel and the slip ratio λ of each tire are important factors that affect the vehicle's performance. The model typically includes the following equations:

- Longitudinal and lateral vehicle speeds:

The longitudinal and lateral speeds, denoted as v_x and v_y , respectively, are often used to describe the vehicle's motion. These speeds can be calculated using the yaw rate, r , and the vehicle's orientation angle, θ , as follows:

$$v_x = r * \cos(\theta)$$

$$v_y = r * \sin(\theta)$$

- Wheel angular speed and tire slip ratio:

The angular speed of each wheel, denoted as ω_{ij} , is a critical parameter in vehicle dynamics models. The individual tire slip ratio, λ_{ij} , is another important factor that describes the tire's deformation during motion. The slip ratio is defined as the difference between the wheel speed and the vehicle speed divided by the vehicle speed. The slip ratio can be expressed as:

$$\lambda_{ij} = \left(\frac{Req * \omega}{v_x} - v_x \right)$$

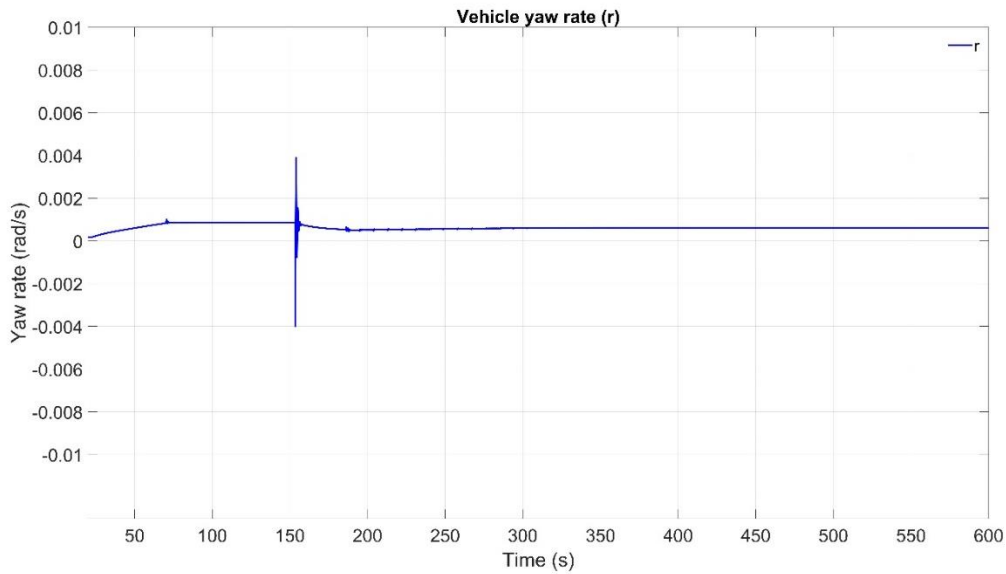


Fig.2.3.1 Vehicle yaw rate

where Req is the wheel equivalent rolling radius, which is a function of the tire radius, vehicle mass, and other factors. The slip ratio affects the tire force and the vehicle's stability during motion. Overall, a vehicle dynamics model that includes

these parameters can be used to predict the vehicle's behavior during various driving scenarios, such as cornering, braking, and acceleration.

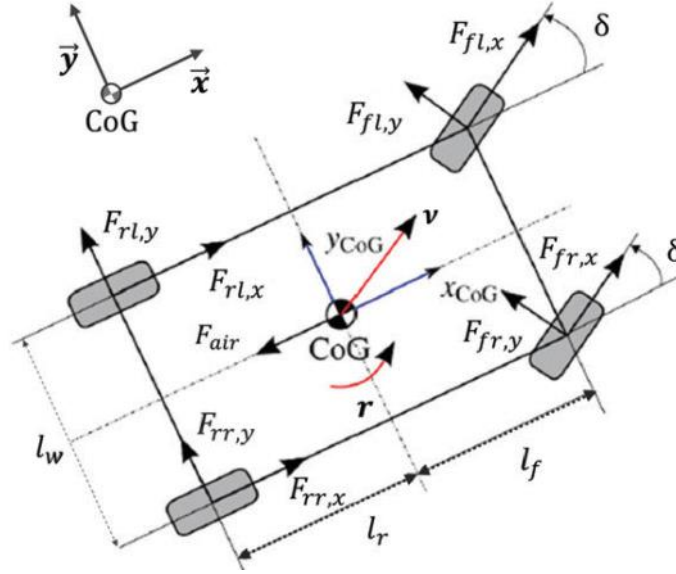


Fig.2.3.2 Full track vehicle planar model

The steering angle of the front wheels is represented by the symbol δ . The inertial properties of the vehicle, including its mass (m), total moment of inertia (J_z) and the total moment of inertia of the wheels (J_t) are also important factors in the vehicle dynamics model. The longitudinal and lateral tire forces, denoted by $F_{ij,x}(\lambda_{ij})$ and $F_{ij,y}$ respectively, are nonlinear and depend on the tire slip ratio (λ_{ij}). In the case where the vehicle is traveling in almost straight-line motion with a high cornering radius and small steering angle ($\delta \approx 0$), the tire forces can be modeled using the Pacejka Magic Formula, which is a semi-empirical model that describes the relationship between tire forces and the tire slip ratio in pure longitudinal slip. In this scenario, the lateral tire forces may be neglected for control design purposes.

$$F_{ij,x}(\lambda_{ij}) = D_{ij,x} \sin(C_{ij,x} \arctan(B_{ij,x}(1 - E_{ij,x})\lambda_{ij} + E_{ij,x} \arctan(B_{ij,x}\lambda_{ij}))) \dots$$

The traction forces, denoted as $F_{ij,x}(\lambda_{ij})$ and hereafter referred to as $F_p(\lambda_{ij})$, depend on the slip ratio λ_{ij} , which can be estimated from measurements of the angular velocities and vehicle speed. The tire parameters $C_{ij,x}$ and $E_{ij,x}$ are constants, while

the parameters $B_{ij,x}$ and $D_{ij,x} = \mu_{ij,x} F_{ij,z}$ account for the tire-road friction and the changing vertical load conditions, which can vary for each tire and are referred to as "road-and-load" conditions..

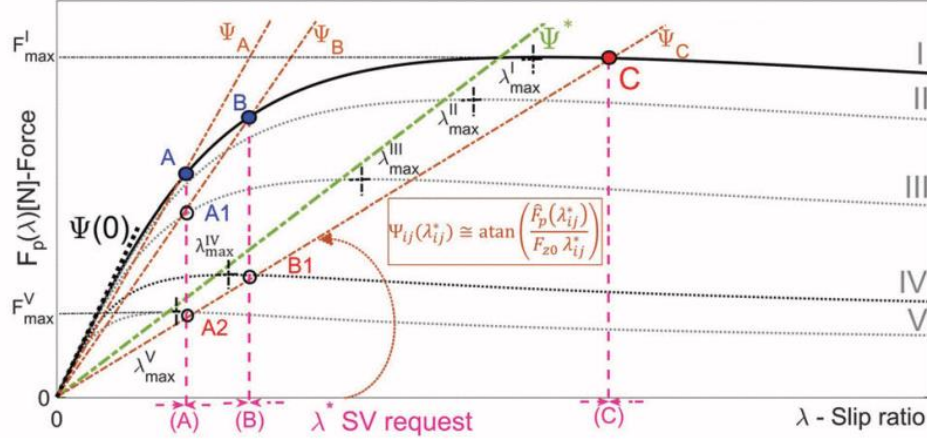


Fig.2.3.3 Pacejka tire force characteristics

The interaction between the control input u_{ij} and the actual forces delivered on the road is captured by the wheel load torque, T_{Lij} , which is linked to the traction force $F_{ij,x}$ through Req figure 2.3.3. The resulting forces are expressed in terms of two terms: f_L and f_D . These terms represent the effects of wheel coordination and enter the vehicle dynamics model as direct control inputs on the longitudinal speed and yaw rate dynamics, respectively. Overall, this model provides a way to capture the complex interplay between the vehicle dynamics, tire properties, and control inputs, which is essential for understanding and designing control strategies for electric vehicles with in-wheel motors.

$$\begin{cases} f_L(\vec{\lambda}) \doteq (F_{rr,x}(\lambda_{rr}) + F_{rl,x}(\lambda_{rl})) \\ f_D(\vec{\lambda}) \doteq (F_{rr,x}(\lambda_{rr}) - F_{rl,x}(\lambda_{rl})) \end{cases}$$

where: $\vec{\lambda} = [\lambda_{rl}, \lambda_{rr}]^T$ is the vector of rear tire-slip ratios; $f_L(\vec{\lambda})$ represents the overall driving traction force including longitudinal forces by rear driven wheels; $\frac{lw}{2} f_D(\vec{\lambda})$ denotes the overall driving yaw-moment, through $f_D(\vec{\lambda})$.

2.4) Model Simplifications

The assumption mentioned above, which approximates the relationship between the total driving force f_m and the differential force f_d using a positive real b and a suitable function $a(\cdot)$, provides a significant advantage in terms of control design. By making this assumption, the control design process becomes simplified and more efficient, as it eliminates the need for lateral velocity measurements and precise model knowledge.

This simplification in control design is particularly useful in situations where exact model knowledge may be difficult to obtain, or where the lateral velocity cannot be directly measured due to limitations in sensing technology or other constraints. The use of this simplified control scheme also results in reduced computational complexity, allowing for real-time implementation and faster response times.

In summary, the assumption that the total driving force and differential force follow certain relationships enables the creation of a control scheme that is independent of lateral velocity measurements and exact model knowledge. This simplifies the control design process, reduces computational complexity, and allows for real-time implementation, making it a useful approach in many practical applications.

$$f_m = a(v_x, v_y)(\omega_{r1} + \omega_{r2}) - b$$

$$f_d = a(v_x, v_y)(\omega_{r1} - \omega_{r2})$$

Physically speaking, this is about what is meant by,

relatively small λ_{r1} and λ_{r2} ,

a sufficiently large curvature radius of the bend maneuver with approximately equal and positive resulting wheel speeds v_{r1} and v_{r2} , namely v_r ,

approximately equal wheel radii $R_{\omega r1}$ and $R_{\omega r2}$,

approximately equal products $D_{lr1}C_{lr1}B_{lr1}$ and $D_{lr2}C_{lr2}B_{lr2}$.

The above approximations, which are somehow rather restrictive and lead to

$$\begin{aligned} a(v_x, v_y) &= \frac{D_l C_l B_l R_\omega}{v_r(v_x, v_y)} \\ b &= 2D_l C_l B_l, \end{aligned}$$

are here introduced at the design stage with the aim of obtaining a theoretically-based, proportional–integral decoupled controller on the motor speeds,⁴ that reduces to the one in Marino, Scalzi et al. (2013) when the two in-wheel motors collapse into one motor and straight manoeuvres are considered.

The approximations mentioned here can be considered as a dynamic equivalent of the $g_\mu(\cdot)$ -existence condition discussed in above section, specifically for electric vehicles that have centralized motors. The simplified system model takes into account the lumped nature of the equations mentioned earlier, and the uncertainties in this model are directly associated with the uncertain nature of the static function $g_\mu(\cdot)$.

2.5) Pacejka Curve

The Pacejka curve, also known as the Magic Formula in the field of tire and vehicle dynamics, is a mathematical model. It was created by Dutch engineer Hans Pacejka and is widely used in the automotive industry for vehicle design and testing, as well as tire performance optimization. The formula is a complex function that considers several factors that affect tire behavior, such as load, speed, and slip angle. The model predicts how a tire will behave under various conditions, such as lateral and longitudinal forces, tire deformation, and slip ratio.

The Pacejka curve Figure 2.5.1 is an important component in vehicle design and development because it assists engineers in understanding the relationship between tire behavior and vehicle handling. Engineers can accurately predict how a vehicle will respond to various driving conditions by incorporating the Pacejka curve into vehicle simulation models and making improvements accordingly. This

allows for tire and vehicle performance optimization, ensuring that vehicles are safe and provide a smooth and comfortable driving experience.

The Pacejka curve is used in motorsports to improve the performance of racing tires, in addition to its use in the automotive industry. Teams can optimize their vehicle setup and improve lap times by accurately predicting tire behavior. The formula has proven to be a valuable tool for tire and vehicle engineers, allowing for the development of vehicles that are safer, more efficient, and perform better. It is widely regarded as one of the most significant contributions to the field of tire and vehicle dynamics, and it continues to play an important role in the industry's advancement. The generic Magic Formula has been developed using mathematical functions that related.

- The lateral force F_y as a function of slip angle α .
- The aligning moment M_z as a function of slip angle α .
- The longitudinal force F_x as a function of longitudinal slip K .

When these curves are obtained from steady state tire testing and plotted, the general shape of the curves is similar to that indicated in figure below. It is important to note that the data used to generate the tire model is obtained from steady state testing. The lateral force F_y and the aligning moment M_z are measured during pure cornering, i.e., cornering without braking, and the longitudinal braking force during pure braking, i.e., braking without cornering.

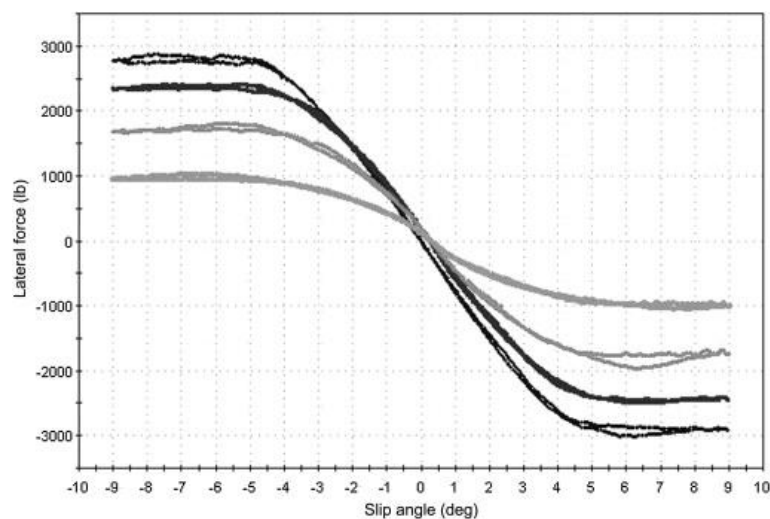


Fig.2.5.1 Pacejka curve example

The basis of this model is that tire force and moment curves obtained under pure slip conditions look like sine functions that have been modified by introducing an arctangent function to 'stretch' the slip values on the x-axis, as shown in the above figure.

$$Y(X) = D \sin (C \arctan(B(1 - E)(X + S_h)) + E \arctan(B(X + S_h))) + S_v$$

**Although we don't use this formula, it has been added for the purpose of an idea*

As indicated above, $Y(X) = y(x) + S_v$, $x = X + S_h$. Whereas, S_h = horizontal shift and S_v = Vertical shift. Y can be either the longitudinal or lateral tire force. When Y is the longitudinal force, X is the slip ratio, and when Y is the lateral force, X is the slip angle. The parameters fit from the experimental data are B , C , D , E , S_h , and S_v .

It is critical to have a realistic tire model that accounts for the slip phenomenon, especially if the car is to be driven at its handling limits and used in a racing context. However, due to the model's complexity, fitting a physical tire behavior to its corresponding parameter value in a Pacejka model can be difficult. However, because the car is used for normal driving in this thesis, a linearized tire model is used, as explained in the following section. This essay provides a unique solution to the structural issue discussed above. It develops a new technique that automatically modifies longitudinal velocity via a clever application of Pacejka curves (see Figure below) (and yaw rate). This approach uses the relevant converging load torque predictions that the motor speed controls provide and is based on the most recent definition of wheel angle. The coordinate pair on the tire force-slip characteristics curve was used to calculate the wheel angle using concurrent force and slip data: The force/slip plane's origin and a point on it that represents a steady-state operation can be found by measuring the angle of a straight line's deviation; this angle contains information about the region in which the tire is operating. Nevertheless, this information is only useful after it is known that a certain characteristic curve will occur (in the absence of which, a cautious limit angle must be determined). The automatic speed adjustment method described in does not even ensure convergence to the specified operating points.

By again using the contraction mapping algorithm on the new suitable ωL -torque current plane, the tire operating point is made to automatically lie in the linear (positive derivative) region of the torque-slip characteristics, next to the maximum,

in this case, in contrast to the wheel angle approach of, where it is made to lie in the region of the torque-slip characteristics near the maximum. Again, no lateral velocity measurements are necessary, and the benefits of the earlier methods—arising from the direct specification of appropriate motor speed references—are maintained. In that order, they are:

- i. motor speed control algorithms that can be directly inherited from the literature on electric machine control; and
- ii. the incorporation of inherently reliable motor speed control loops.

Comparative CarSim simulations are included to demonstrate the efficacy of the proposed approach in the presence of uncertain parameters and complex vehicle dynamics - such as aerodynamic components, torsional and elastic dynamics, pitch, roll, and vertical displacement dynamics - that were overlooked during the control design stage.

3) Control Scheme

The process of creating a system or device that governs or controls the behavior of another system or process is known as control design. It involves modifying a process's input or output such that it behaves in a desired way by using mathematical models, algorithms, and physical components. Several fields, including robotics, aircraft, automotive, chemical, and electrical engineering, require control design. The ultimate objective of control design is to guarantee that the controlled system performs with the fewest errors and best disturbance rejection possible. The architecture can be decomposed as,

System modeling and Controller design: This involves creating a mathematical model of the system to be controlled. The model includes the system's input, output, and internal dynamics. This involves designing a controller that takes the system's input and output into account and generates a control signal that can manipulate the system's behavior.

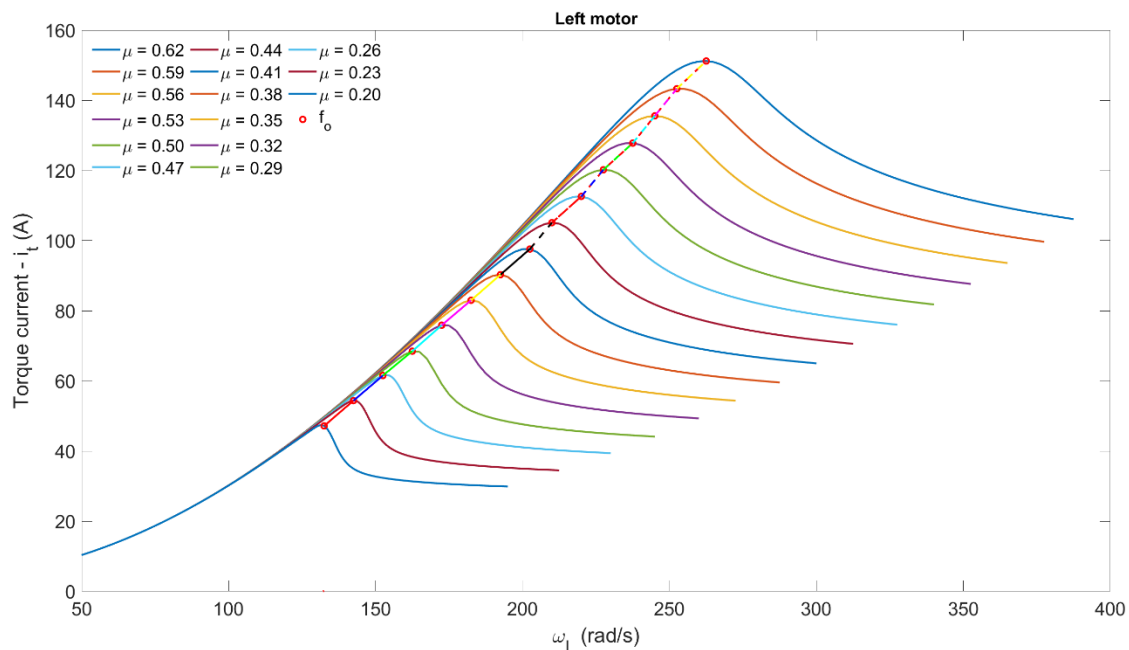


Fig.3.1 Left motor curve

Reference signals are provided by the Reference Generator. It enables the computation of the reference speed profile as well as the geometric trajectory that specifies the path to be taken. Control makes use of these two various reference

signals that were computed at this level. And by delivering the proper control signals, in this case the acceleration, deceleration, and steering angle of the front wheel, the Control assures the automatic vehicle guidance along the created trajectories. For effective vehicle guidance, simultaneous longitudinal and lateral control is necessary.

In order to demonstrate the effectiveness of the suggested approach in the presence of uncertain parameters and complex vehicle dynamics that have been overlooked at the control design stage, such as aerodynamic components, torsional and elastic dynamics, pitch, roll, and vertical displacement dynamics, comparative CarSim simulations are included.

3.1) Control design

In a bend maneuver with (sufficiently large) continuous curvature radius, the issue of managing both the longitudinal speed and the yaw rate must be resolved (straight maneuvers are allowed as a degenerative case),

$$\omega_L = \frac{\omega_{r1} + \omega_{r2}}{2}$$
$$\omega_D = \omega_{r2} - \omega_{r1}$$

These signify, respectively, the longitudinal and differential parts of the wheel speeds. Hence, using the aforementioned equations enables the development of the following dynamic equations.

$$\begin{aligned}
\dot{v}_x &= rv_y + \frac{1}{m} [-(f_{sf1} + f_{sf2})\sin(\delta) - c_a v_x^2] \\
&\quad + \frac{1}{m} (2a(v_x, v_y)\omega_L - b) \\
\dot{v}_y &= -rv_x + \frac{1}{m} [(f_{sf1} + f_{sf2})\cos(\delta) + f_{sr1} + f_{sr2}] \\
\dot{r} &= \frac{l_f}{J_z} (f_{sf1} + f_{sf2})\cos(\delta) - \frac{l_r}{J_z} (f_{sr1} + f_{sr2}) \\
&\quad + \frac{l_w}{2J_z} (f_{sf1} - f_{sf2})\sin(\delta) + \frac{1}{J_z} \sum_{j \in \{f1, f2, r1, r2\}} M_{zj} \\
&\quad + \frac{l_w}{2J_z} a(v_x, v_y)\omega_D,
\end{aligned}$$

This emphasizes the input-output decoupling properties of ω_L and ω_D , which are inherited from the definitions of f_m and f_d as well as from the expressions of i . Hence, the regulation of v_x and r are handled separately by the control inputs ω_L and ω_D . Additionally, because the transformation is invertible, ω_L and ω_D unambiguously specify the control inputs ω_{r1} and ω_{r2} .

$$r_* = \left(\frac{\delta}{l_f + l_r} v_{x*} \right)$$

The constant reference for the yaw rate r_* is mentioned above, it is defined in terms of the constant reference v_{x*} for the longitudinal velocity v_x .

To account for these uncertainties, the control system uses feedback from sensors to compute the error signals $\tilde{v}_x$ and $\tilde{r}$, which are the discrepancies between the desired reference values and the actual ones. The proportional-integral (PI) controller adjusts the control actions and reduces the error using these error signals as inputs. The control system may control the vehicle's longitudinal speed and yaw rate to maintain a stable and secure operation by continuously modifying the control inputs depending on the error signals. It is crucial to remember that the quality of the sensor measurements and the fine-tuning of the controller gains determine how accurate the control system is. To ensure that the system responds

fast and smoothly to changes in the reference inputs and disturbance inputs while preserving stability, the controller gains must be effectively calibrated. In addition, the system's ability to precisely predict the vehicle's motion and direction depends on the choice of the right sensors and where they are installed. Hence, the proportional–integral control is designed accordingly,

3.2) Control structure

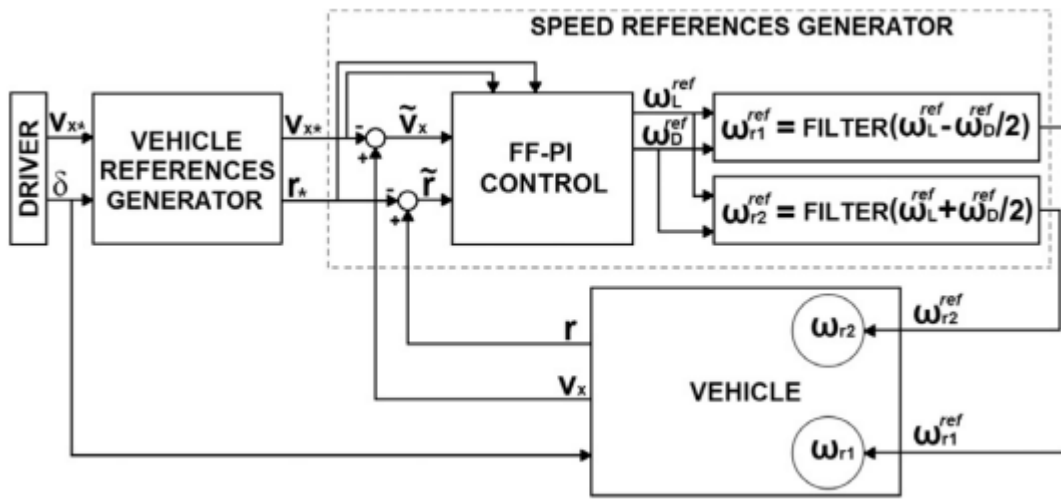


Fig.3.2.1 Block diagram for the control architecture involving.

The figure 3.2.1 provided above depicts the entire control system architecture described in this section, illustrating the key components and their related signal flows. The driver specifies the desired constant longitudinal speed, denoted as v_{x*} , and chooses the constant steering angle, δ . For autonomous vehicles, the proposed control strategy can be directly implemented by acquiring these reference values through external devices. The desired yaw rate, r_* , is supplied by another component. Using a combination of feed-forward and feedback error actions, the control system generates reference signals for the rotor speeds of each electric motor in accordance with the desired outputs. However, to make their time derivatives well-defined and available for feedback, these reference signals undergo filtering using stable linear filters with unitary gain, characterized by a

positive design parameter $\lambda_{\omega,i}$ for each electric motor rotor speed reference signal, denoted as ω_{r1} and ω_{r2} .

The use of stable linear filters in the control system is important to ensure that the reference signals for the rotor speeds of each electric motor are well-defined and can be utilized in feedback loops. These filters are designed using positive parameters, denoted by $\lambda_{\omega,i}$, where $i \in \{\omega_{r1}, \omega_{r2}\}$. The choice of $\lambda_{\omega,i}$ affects the performance and stability of the control system, and its value must be carefully selected. The feed-forward and feedback error actions utilized in the control system allow for accurate tracking of the desired reference signals, which are chosen by the driver for the longitudinal speed and steering angle and provided by another component for the desired yaw rate. Overall, the control system architecture presented in this section provides a comprehensive approach for controlling the motion of electric vehicles, allowing for precise tracking of desired outputs and robust performance in the face of system uncertainties.

$$F_{\omega,i} = \frac{\lambda_{\omega,i}^{p\omega}}{(s + \lambda_{\omega,i})}$$

whose outputs ω^{ref}_{r1} and ω^{ref}_{r2} are the sufficiently smooth references to be incorporated into the motor speed tracking control schemes driving each in-wheel motor. If we set the values of k_{px} , k_{ix} , k_{pr} , and k_{ir} to zero, we can obtain the traditional open-loop configuration. In such a case, ω_D becomes zero when the desired yaw rate r^* is also zero. This means that the lateral controller's actions will not have any impact on the vehicle's movements. The classical open-loop configuration is a simpler system as it does not incorporate feedback to adjust the control actions based on the system's response. However, it also means that it may not be as effective in handling uncertainties and disturbances that can impact the vehicle's performance.

4) Simulations

CarSim is a widely used simulation environment that is designed to accurately model the dynamics of complex vehicular systems. In this particular case, CarSim is used to carry out simulations that illustrate the theoretical principles presented in chapter 2. The simulations are conducted using a comprehensive model of a rear-wheel drive A-class vehicle, which takes into account the complex interactions between various subsystems such as the suspension, steering, tires, and road surface. One of the key features of the CarSim model is the use of detailed, nonlinear tire models that take into account the combined effects of lateral and longitudinal forces. These models are essential for accurately predicting the vehicle's behavior under different driving conditions, such as cornering, braking, and acceleration. Furthermore, the model includes the major kinematics and compliance effects of the suspension and steering systems, which are represented by nonlinear spring models. These components are critical for understanding the vehicle's handling characteristics and response to driver inputs.

In addition to the vehicle's mechanical properties, the CarSim model also incorporates the aerodynamic characteristics of the vehicle. This information is vital for predicting the vehicle's behavior at high speeds, and it can help to optimize the vehicle's design for improved efficiency and performance. The use of in-wheel induction motors in the CarSim vehicle model is an interesting choice, as it allows for a greater degree of control over the vehicle's power delivery and torque distribution. This setup is becoming increasingly popular in electric and hybrid vehicles, as it can lead to improved efficiency and handling. Overall, the CarSim model presented in this work is a highly detailed and accurate representation of a rear-wheel drive A-class vehicle, and it provides valuable insights into the vehicle's behavior under different driving conditions. Such models can be used to optimize vehicle designs and inform decision-making in the automotive industry, and they are an important tool for advancing the field of vehicle dynamics. Compared to internal combustion engines, induction motors offer several advantages when power levels are comparable. One of the key benefits is their ability to respond rapidly to changes in torque, which is between 10 to 100 times faster than internal

combustion engines. This makes induction motors ideal for applications that require dynamic torque adjustments, such as electric traction. Another important advantage of induction motors is their regenerative braking capability. This feature enables the motors to recover energy during braking, which can be used to power other vehicle systems. This makes induction motors an eco-friendly choice for green and energy-efficient electric traction applications.

4.1) CarSim Adjustments

Before sending the data to Simulink from carsim there are few data that should be seen, from one of the many changes, one of the changes that need to be done is setting the Math model Time step to 0.00002 (see figure 4.1.1). The passage explains how output variables are written into a binary file and associated with a VS or ERD file that contains header information needed to read the binary file. It also describes how the default time steps for the numerical integration and output file can be specified in different places within the database. To reduce the size of output files, the user can choose to write outputs at some multiple of the model time step. The default time steps can be set in the Preferences screen, which will be used if no other values are specified in the datasets used for the simulation. Alternatively, time step values can be set on all Model screens, including the one used to specify a Simulink model. This can be necessary when using an external model that requires a specific time step that is not standard for other runs. The time steps specified in a linked Model dataset will override the values from the Preferences dataset. From the above, time step values can also be set on the Run Control screen, which will override any other values. To set the time step, the user can check the "Set time step here" box, which will display data fields for setting the time steps for the numerical integration and output file. However, this control is only visible when "Show more options on this screen" is checked.

☒ Set time step here

	Time step (s)	Freq. (Hz)
Math model:	2e-05	50000
Output file:	0.02496	40.0641

Do not set time, station, or direction here ▼

Fig.4.1.1 Time step section in Carsim

The time step can be specified using either seconds or frequency, and the browser automatically maintains the correct inverse relationship between these two values. If the user modifies the time step, the frequency is calculated and updated, and if the frequency is modified, the time step is calculated and updated. The browser also maintains the correct inverse relationship between time step and frequency for the output file and ensures that these numbers are multiples of the internal time step used in the VS math model.

The next one of the important part of starting with carsim is uploading the MATLAB/Simulink files in working directory of carsim and modifying the import and export channels before we hit run or sending the data to Simulink for feedback control and modifying the error. This passage provides instructions for specifying an external model or program file to be used with the VS Solvers. The model or program can be a Simulink file and the user can provide the pathname to the MDL or SLX file using the adjacent browse button or by typing it in.

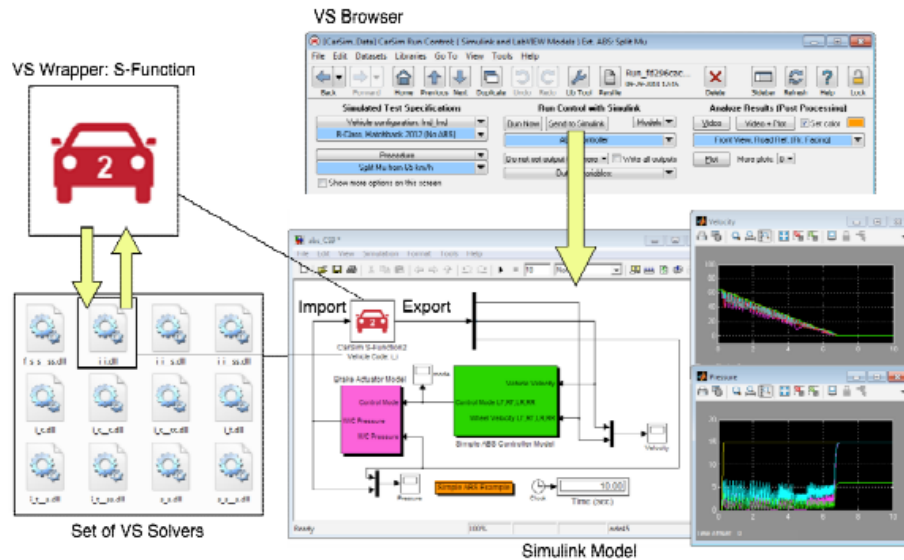


Fig.4.1.2 CarSim and Simulink

In Simulink, if the user wishes to use a working directory that is different from the active VS database folder (e.g., CarSim Data), they can check the adjacent checkbox to activate the Simulink working directory field. The pathnames shown on the screen will be relative to the selected working directory. The passage also mentions the option to specify alternative solver programs for multiple vehicle configurations. In this case, the DLL files need to have different names in order for the vehicles to be independent. The user can rename the original DLL file to create distinct names for each solver program.

After creating a Omega L reference, and identifying the time for the simulation to perform. In our case we have the ramp-to-ramp omega L reference to almost 600rad/sec for which the time to run the simulation for entire reference model is nearly equal to 3500 sec, we can modify the simulation time before sending it to Simulink in carsim in section start and stop conditions. data fields are shown for the start time (keyword = TSTART) and starting station (keyword = SSTART) along the path used by the driver model. The direction the vehicle is travelling relative to the path is determined either by setting it directly with a drop-down control, or by comparing start and stop station values.

Now, after setting a start and stop conditions, we have to design the road path according to our willing in the section Reference path segment. A VS Reference Path is a continuous line that is situated in a horizontal plane and has a smooth

gradient without sharp corners. The primary purpose of a reference path is to establish a 2D coordinate system for describing locations near a path of interest. The path coordinates are the station S (distance along the path) and lateral coordinate L (distance a point is from the path, measured on a line that intersects the point and the path, and is perpendicular to the path at the point of intersection). The S coordinate is based on an axis that is fixed, while the L coordinate is based on an axis whose location and direction are variable. The S and L coordinates associated with a path have a wide range of applications in VS Math Models. Closed-loop steering controllers use the absolute value of the L coordinates of preview points ahead of the vehicle relative to a target path to operate. Objects such as traffic vehicles, targets for on-board sensors, etc. can be located relative to the path using S and L . Lanes are defined using constant values of L for a Reference Path, or possibly, with the Configurable Function LTARG that defines L as a function of S . Road surface elevation and friction are defined as functions of S and L for a road reference path.

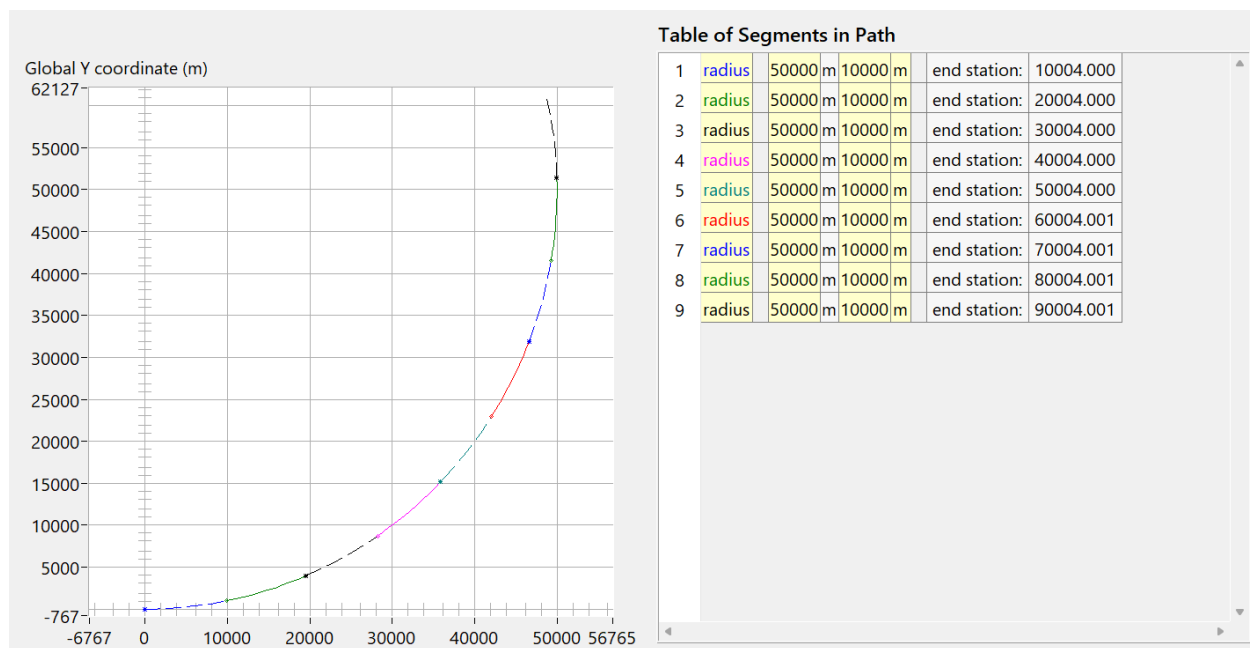


Fig.4.1.3 Road segment section in CarSim

In our case the road segment is arc-shaped (see figure 4.1.3) and entire reference is made with up to 9 semi segments with almost radius of 50000 and 10000 with global X and Y coordinates. Transforming between S - L and X - Y coordinates in VS

Reference Paths is a critical aspect of the modeling process. While converting from S-L to X-Y coordinates is a straightforward process that does not require iteration, the reverse transformation requires an iterative approach because the segment containing the X-Y pair must first be located. This iterative process is necessary to determine road properties where tire/ground contacts occur, and to determine the relationship between points of interest and a target path for steering and speed controllers in the vehicle model. VS Solvers use instances to keep track of the segment on which an object is located and utilize this information to quickly locate the correct segment. However, a significant complication arises when multiple pairs of S-L coordinates would be valid for a single point of interest. This is because a line perpendicular to the reference path can be drawn from any point on the path to the point of interest, resulting in different but equally valid S-L coordinates. Therefore, it is important to select the correct S-L coordinates for a point of interest, and in the case of a racetrack, for example, point A is typically the desired location on the path.

After changing the road reference segment, we can visualize and modify the road and can also the friction to the road. To change the road visuals, we can do it in Update road surface section, Besides the roles played in building the simulation model, the road information can be used to generate 3D shapes that provide video visualization. Advanced users sometimes define multiple roads with geometric shapes that will not be used with the vehicle model, but which will be used to generate path-related shapes such as roads in the background, curbs, walls, stripes on pavements, etc. In these cases, a single road can be used multiple times to generate different shapes. The 3D Road Surface Viewer in VS Visualizer is used to create and control the appearance of animator shapes used to view 3D road surfaces. This screen allows you to define up to 25 3D shape files in the form of rectangular "lanes" that follow the reference line of the road. These lanes are created using S-L road coordinates and will curve as needed to accurately depict the road's geometry. The dimensions of these rectangular lanes can be specified on the screen to control the level of detail displayed in the visual representation of the road. To create stripes and grids, the rectangles can be connected or alternated.

MTL File that defines textures: Roads\materials\road.mtl ☐ Mirror texture images Optimization Threshold: mm

	Color	Color	Material Type	Spec.	S Scale	L Tiles	L Start	Units	L Stop	Units	Div's	i Use	S Start	S Stop	Interv
1	1 1 1		Road (Rough)		24	1	-10	m	10	m	1	1	-50	90000	5
2	1 1 1		Grass (Dark)		24	1	-10	m	-100	m	1	1	-50	90000	5
3	1 1 1		Water		24	1	10	m	150	m	1	1	-50	90000	5

Fig.4.1.4 Visuals Table

The datasets associated with this screen are used by VS Road Calculator, which automatically runs whenever a change is made to a dataset in the Road: 3D Surface library or when the Update Road Surface 3D Shapes button is clicked on the Road: 3D Surface screen. This tool provides detailed information about the 3D geometry of the road, including road surface elevations and friction, which can be used in simulations for various purposes such as testing of vehicle models or traffic flow analysis. By using the 3D Road Surface Viewer in conjunction with VS Road Calculator, users can generate highly accurate visual representations of the road surface for a variety of applications.



Fig.4.1.5 Carsim visuals

The above figure 4.1.5 representing Water on left side, and dark green grass on right side and the road is represented as tar rough road. Which will act as our main road.

Now, after changing the Road segments and road visuals, let's take a peek into the road adhesion coefficient, one of the most important parts to take place in MATLAB simulation.

4.2) Adhesion Coefficient

The coefficient of adhesion is a critical parameter in the field of transportation engineering. It is a fundamental value that characterizes the interaction of a vehicle's wheels with the road surface, which has a direct impact on road safety. The coefficient of adhesion is used to evaluate the performance of various types of road surfaces, including pavements, asphalt, and concrete. By analyzing the maximum tangential force acting along the road and the normal reaction of the wheel in the contact area, transportation engineers can determine the coefficient of adhesion.

One of the most important uses of the coefficient of adhesion is to determine the safe speed of a vehicle on a particular road surface. For example, if a road surface has a low coefficient of adhesion, vehicles may have a tendency to skid or slide. In such cases, it may be necessary to reduce the speed limit or implement additional safety measures to prevent accidents. Automatic speed reduction systems, such as those in modern vehicles, can detect abrupt changes in the coefficient of adhesion and adjust the vehicle's speed accordingly. The coefficient of adhesion is a critical factor in road safety and plays a significant role in designing safe and efficient transportation systems.

The vehicle speed can be thus automatically reduced when an abrupt change in tire–road adhesion coefficients occur.

NO.	TYPE AND CONDITION OF THE ROAD	ADHESION COEFFICIENT (μ)
1	Concrete Dry	0.7-0.8
2	Concrete Wet	0.5-0.6

3	Dry Dirt Road (link)	0.5-0.6
4	Wet Sand	0.4-0.5
5	Dry Loam (link)	0.4-0.5
6	Compacted Snow (link)	0.3-0.5
7	Wet Dirt Road (link)	0.2-0.4
8	Concrete - covered with up to 5cm of snow	0.2-0.4
9	Loose Snow (link)	0.2-0.4
10	Dry Sand	0.2-0.3
11	Meadow covered with turf (link)	0.1-0.4
12	Loam – moistened to fluid state	0.15-0.25
13	Smooth Ice (Temperature below 0°C) (link)	0.05-0.1

*All the data of adhesion coefficient values are taken by a research paper published by Saint Petersburg State University of Architecture and Civil Engineering. The link to the research paper is pasted [here](#).

In the first step of our simulations, we need to take friction values from above table. The changing of adhesion coefficient values can be found in the carsim section called friction. From this section we can change the adhesion coefficient values accordingly, it can be constant for entire road segment, or it can also be split into different mu vales across the road segment. The passage discusses two libraries in CarSim that can be used to specify the friction between a tire and the road surface as a function of the S-L coordinates of the path. These libraries are Road: Friction Map, S-L Grid and Road: Friction Map, Variable Width. The former is used when a rectangular grid of S-L coordinates is sufficient, while the latter is used for roads with varying widths or frictional changes that are not parallel or perpendicular to the road's reference line.

The tire models in CarSim support two coefficients of friction - one for lateral forces and aligning moment and another for longitudinal force. While advanced users can import separate coefficients, the coefficient of friction is typically specified as a function of S and L. The coefficient of friction is one of the main parameters that characterize the performance of road surfaces and the interaction of the tire with the road. Its value is used to determine the safe speed of a vehicle. Physically,

coefficients of friction between a tire and the ground are always positive, but the VS Math Models allow zero and negative values to be specified.

A friction value of zero will result in the tire models producing zero shear force and moment. This is useful in special applications such as kinematic and compliance (K&C) test rigs where tire forces are disabled with bearing plates. The Road: Friction Map libraries and the configurable function controls in CarSim provide a way to accurately model the interaction of a tire with the road surface based on the coefficient of friction. By specifying the friction as a function of S and L coordinates, the tire models in CarSim can be used to determine the safe speed of a vehicle and its performance on different road surfaces. If friction is set to a positive value, then that value (MU_ROAD) is used for both the lateral and longitudinal friction level in the internal tire model.

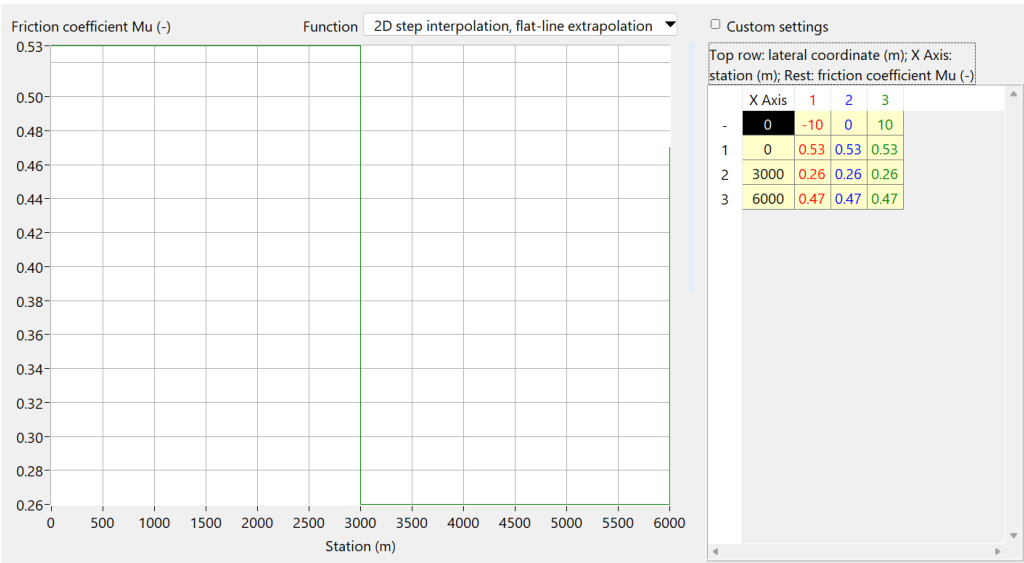


Fig.4.2.1 Adhesion Coefficient section

After modifying these important data now, we need to prepare to send the data to MATLAB/Simulink for performing simulation.

While making simulation we can also see the how the car is moving on the road segment. We can also see the graphs of different notations simultaneously with the video visuals. The user can control the playback of the simulation and adjust the speed, start and stop times, and view different angles of the vehicle. The visualizer

also allows for zooming in and out of the scene, panning across the landscape, and highlighting specific parts of the vehicle for closer inspection.

The combination of real-time 3D graphics and simulation data provides a powerful tool for engineers and researchers to visualize and analyze the behavior of their vehicles under various conditions. By observing the visual feedback, users can gain a deeper understanding of how their designs behave in different scenarios and identify areas for improvement. Overall, the visualizer is an essential component of the simulation toolkit, providing users with a highly effective way to analyze and optimize vehicle performance.

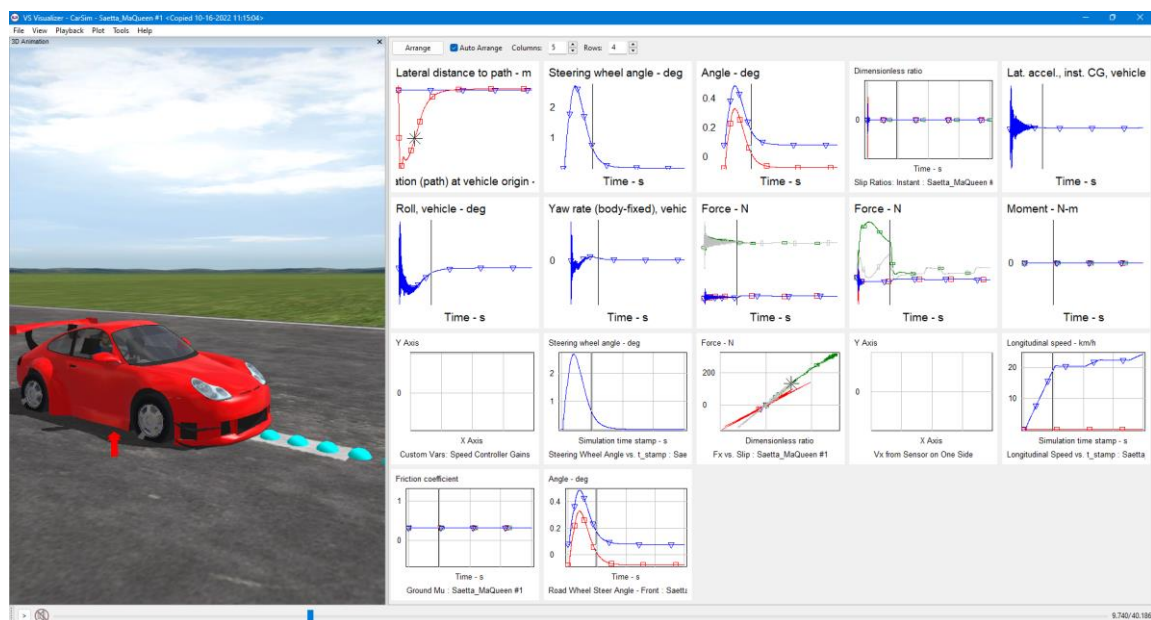


Fig.4.2.2 Simulation visuals with graphs

The communication between CarSim and other software occurs using import and export variables. The math model equations of CarSim include the effects of variables that are defined in other software and imported during the simulation. There are around 400 import variables and 800 output variables available in CarSim, and you can define more using VS Commands.

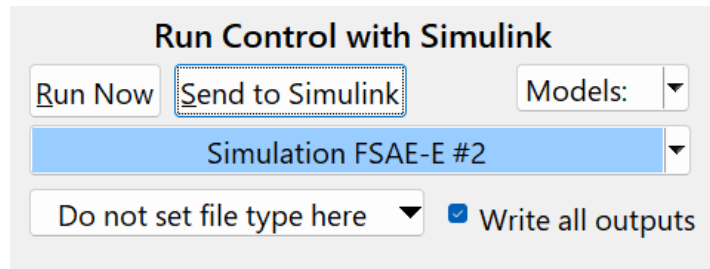


Fig.4.2.3 Run Simulink from carsim

Simulink is a popular graphical user interface for building math models as block diagrams for a variety of fields, including control systems. The VS Browser simplifies the connection between CarSim and Simulink. You can work in the VS Browser and use Simulink models in the same way as built-in math models. You can also use CarSim's vehicle dynamics models from within Simulink by clicking the Send to Simulink button, transferring all necessary data to Simulink, and using the S-function plug-ins.

While CarSim vehicle models have hundreds of available import and export variables, you don't need to handle all of them when using the model in Simulink. Instead, you can activate only the necessary variables by pointing-and-clicking from the VS Browser screens. Furthermore, you can transfer back to the VS Browser at any time to adjust vehicle properties or use VS Visualizer to view the results generated in Simulink.

4.3) MATLAB/Simulink models

Before beginning simulations in Simulink, it is necessary to run a few data scripts in MATLAB. These scripts allow Simulink to recognize the specific CarSim models being used in the simulation. One of the scripts provides all the data about the vehicle's velocity V_x , as well as other parameters obtained from CarSim. This includes information about the vehicle mass, pilot mass, battery mass, wheel mass, iwm mass, steering ratio, velocity mapping, and slip saturation, among other parameters. Another script provides information about motor parameters, derivative parameters, initial conditions, speed controller, PSID controller, and the Eigenvalue assignment of Torque controllers. Finally, a third script contains the information for the Eigenvalue assignment of cruise control and lateral dynamics

and torque controller. The control algorithm used in the simulation includes no feedforward terms and PI gains of (25, 15) and (20, 10) for the Torque and lateral dynamics controllers, respectively.

All of this data is necessary to ensure accurate and precise simulations of the vehicle's performance. The simulation can also be adjusted by using various inputs for the vehicle model, such as the throttle and steering input signals. These inputs can be provided by the user or generated through an automated testing procedure. The output of the simulation includes important performance metrics such as vehicle speed, acceleration, and fuel economy. These metrics can be used to evaluate the performance of the vehicle in various driving conditions and to optimize the design of the vehicle's systems.

In conclusion, the use of data scripts in MATLAB and Simulink allows for accurate and precise simulations of a vehicle's performance. By providing a comprehensive set of data about the vehicle's parameters, the simulation can accurately model the behavior of the vehicle in various driving conditions. This information can then be used to evaluate and optimize the design of the vehicle's systems. Additionally, the use of various input signals in the simulation allows for a thorough analysis of the vehicle's performance under different driving conditions.

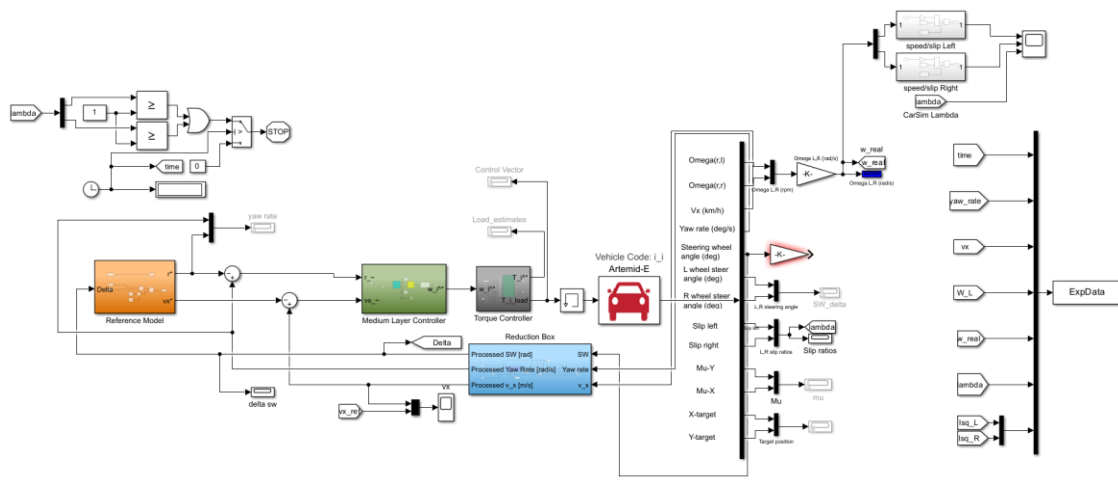


Fig.4.3.1 Full Simulink model

Simulink is a graphical programming environment provided by MATLAB that allows engineers and scientists to model, simulate, and analyze dynamic systems. The Simulink model shown in the image is used to simulate the behavior of a vehicle with varying adhesion coefficients. The adhesion coefficient is a measure of the

traction or grip between the vehicle's tires and the road surface, and it affects the vehicle's ability to accelerate, decelerate, and maintain stability. The Simulink model consists of various blocks that represent different components of the vehicle, including the engine, drivetrain, wheels, and suspension. These blocks are interconnected to create a system that simulates the vehicle's dynamics and behavior. The model also includes a data logging block that records the output of the simulation, which can be saved to an Excel file for further analysis. To perform the simulation, the model is run with different values of the adhesion coefficient (μ) ranging from 0.20 to 0.62 in increments of 0.03. This allows the behavior of the vehicle to be evaluated under different road conditions and adhesion levels. By analyzing the simulation data, engineers can gain insights into the vehicle's performance and make adjustments to improve its handling and stability. Overall, the Simulink model provides a powerful tool for studying the behavior of vehicles and other dynamic systems, and the ability to log simulation data in Excel files enables engineers and scientists to perform detailed analysis and gain insights that can inform design decisions and improvements.

The above Simulink model (from the image) have many different sub models inside medium layer controller, Torque controller, Reference model, Reduction box. Whereas the Artemid-E block is the Solver_SF CarSim solver block that import all the data from carsim to this model. And from there the data enters a multiplexer from where the data is different subsequent blocks.

In Simulink, the Mux block is used to combine multiple signals into a single output signal. It takes multiple input signals and merges them into a single output signal that can be processed or displayed in a subsequent block. The input signals can be of any data type, such as scalar, vector, or matrix, and they can have different dimensions and data types. The Mux block is often used in Simulink models to simplify the model and reduce the number of signal lines between blocks. For example, if there are multiple signals that need to be combined and passed to a subsequent block, rather than connecting each signal to the subsequent block separately, the signals can be connected to a Mux block and then the Mux block can be connected to the subsequent block. This can help to make the model easier to read and understand. The Mux block can also be used to select different input signals based on a control signal or switch between input signals dynamically based on different conditions. The Mux block is a fundamental building block in Simulink,

and its flexibility and versatility make it a useful tool for many applications in Simulink modeling.

In the Simulink model, the Mux block is used to combine multiple signals into a single output signal, which is then passed to the Reduction box block figure 4.3.2. The Reduction box block is an important block in the model that helps to control the error in the simulation. The Reduction box block receives three main signals from the Mux block: the steering wheel angle (sw), the yaw rate, and the velocity (V_x) of the vehicle.

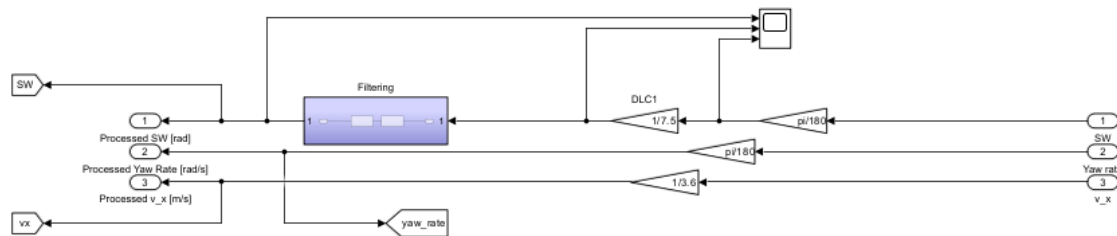


Fig.4.3.2 Reduction Block

These signals are processed within the Reduction box block to calculate the error between the desired and actual vehicle trajectory. The error signal is then passed to other blocks in the model, such as the PID controller and the vehicle dynamics block, to adjust the steering angle and other parameters of the vehicle to minimize the error. The PID controller block receives the error signal from the Reduction box block and uses it to adjust the steering angle of the vehicle. The PID controller block calculates the necessary adjustment based on the error signal and the gains specified in the block parameters. The adjusted steering angle is then passed to the vehicle dynamics block, which simulates the behavior of the vehicle based on the new steering angle and other inputs, such as the vehicle velocity and road conditions. Other blocks in the Simulink model, such as the tire dynamics block and the road profile block, use the signals from the Mux block and other blocks to simulate the behavior of the vehicle under different conditions. By combining and processing these signals, the Simulink model can simulate the behavior of the vehicle in a variety of scenarios and provide insights into how the vehicle will perform in real-world conditions.

The Reference longitudinal speed block is made inside medium layer controller. Reference longitudinal speed, also known as desired or set speed, is the target speed that a vehicle is programmed or instructed to achieve. In the context of vehicle dynamics and control systems, longitudinal speed refers to the speed at which the vehicle is traveling along its longitudinal axis, which is the axis that runs from the front to the rear of the vehicle. In our case the Reference longitudinal speed is built with ramps with difference of 2.5 rad/sec with the total of 600 rad/sec for the time period of around 3500 seconds.

In a vehicle control system, the reference longitudinal speed is often set by the driver or by a control algorithm. The control system then adjusts the vehicle's throttle, braking, and other parameters to maintain the desired speed. The reference speed can be a constant value or can vary depending on the driving conditions, such as the road gradient, traffic, or weather. Reference longitudinal speed is an important parameter in many vehicle control systems, such as adaptive cruise control and automatic emergency braking systems. In these systems, the reference speed is used to determine the distance to the vehicle ahead and to apply the brakes or throttle to maintain a safe distance from the vehicle ahead. The reference speed can be expressed in various units, such as kilometers per hour (km/h), miles per hour (mph), or meters per second (m/s), depending on the convention used in the particular system or application.

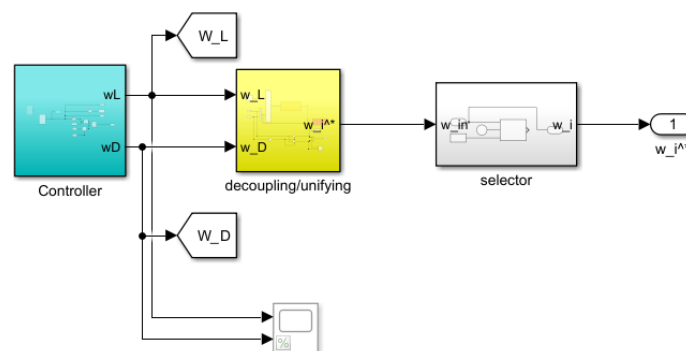


Fig.4.3.3 Controller block

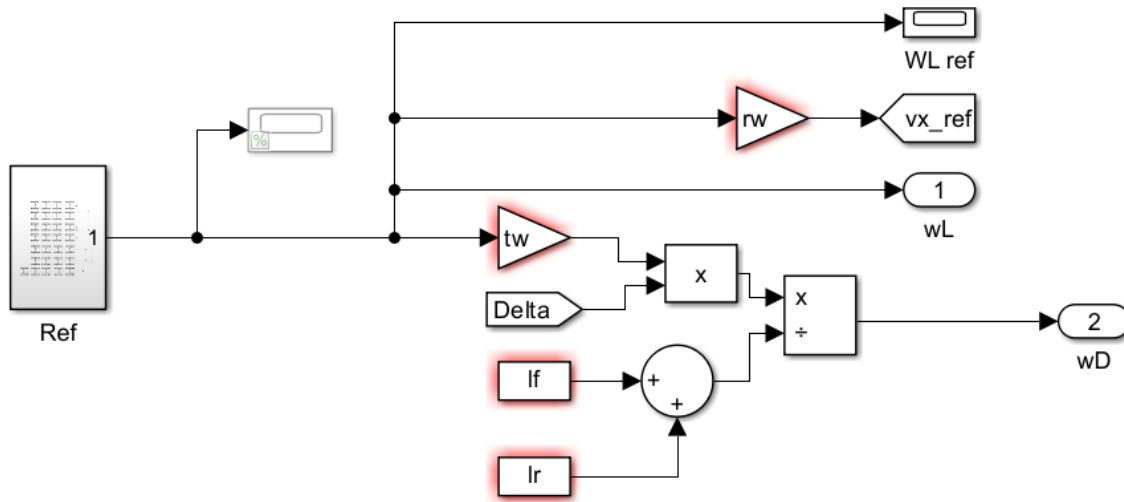


Fig.4.3.4 Longitudinal speed reference block

The Controller block figure 4.3.3 in a Simulink model is responsible for processing inputs and generating outputs to control the behavior of the system. In the context of an electric car, the Controller block is designed to generate two output signals, w_L and w_D , which represent the reference wheel speeds for the left and right wheels of the car. These signals are then fed into the Decoupling block, which uses a decoupling algorithm to ensure that the car's actual motion matches the desired motion specified by the reference wheel speeds. By carefully controlling the wheel speeds, the decoupling algorithm can improve the car's stability, traction, and energy efficiency. Overall, the Controller and Decoupling blocks are critical components of the Simulink model that help to optimize the performance and safety of the electric car. Now, when performing a bend maneuver with a constant curvature radius, it is important to regulate the longitudinal speed to maintain a safe and stable driving condition. This requires a control system that can adjust the throttle and braking to maintain the desired speed. The system must also be able to handle the degenerative case of straight maneuvers.

$$\omega_L = \frac{\omega_{r1} + \omega_{r2}}{2}$$

$$\omega_D = \omega_{r2} - \omega_{r1}$$

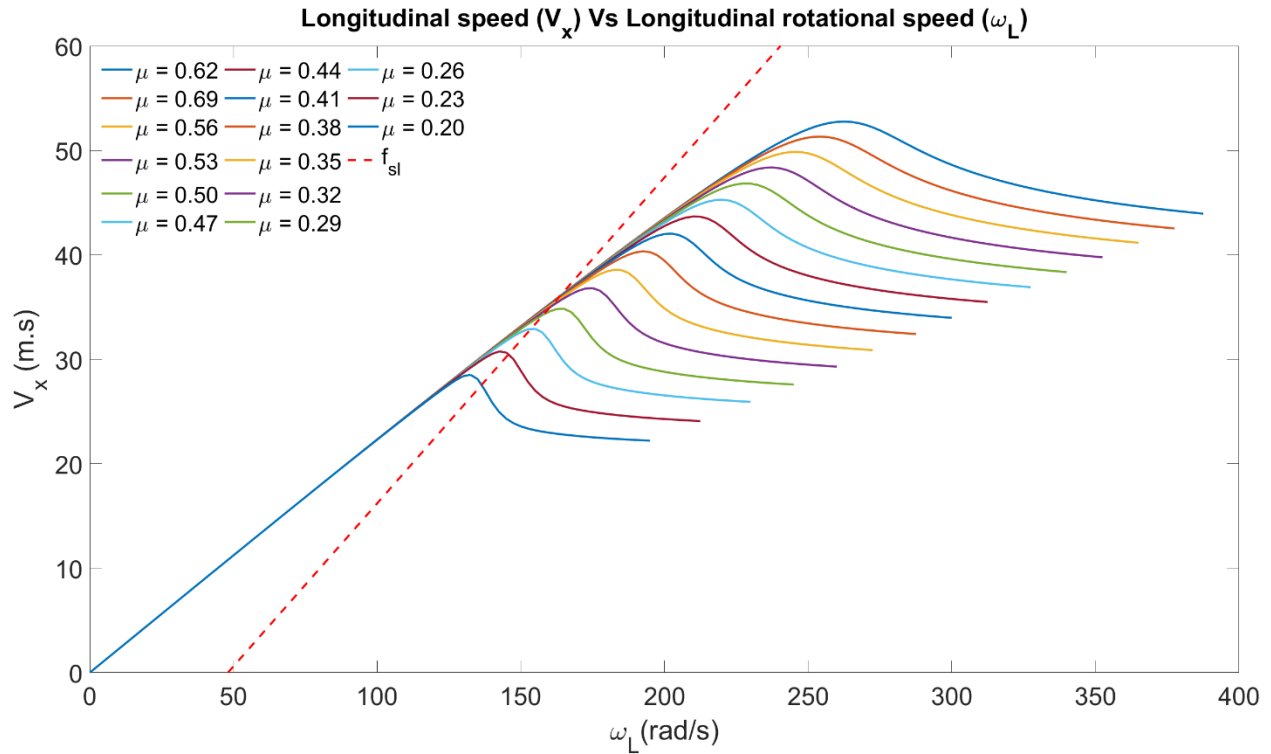


Fig.4.3.5 V_x vs Longitudinal rotational speed

The graph above displays simulation results of the longitudinal speed and longitudinal rotational speed of a model, showcasing 15 different adhesion coefficient values. It is evident that the simulation stopped much earlier for the slippery surface, in contrast to the surface with high friction.

When Slip ratio λ reaches 1,

The slip ratio is similar to slip angle but instead of being measured in angular displacement, it relates the amount of slip a tire experience relative to a sliding condition. To be mentioned, a slip ratio of 0 is a free rolling tire and a ratio of 1 is a tire that has lost traction.

And the Simulink model encouraging the above statement is as follows,

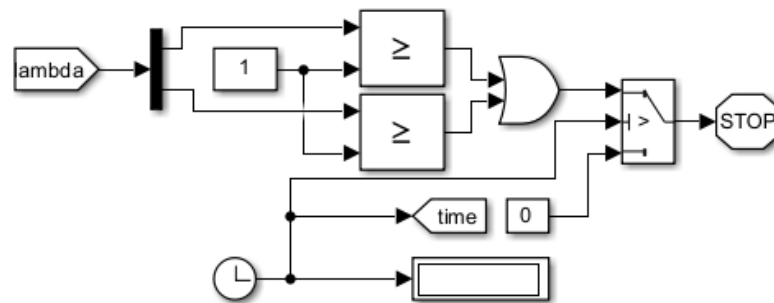


Fig.4.3.6 Slip constant block.

The slip constant λ plays a crucial role in determining the stability and performance of the vehicle during a maneuver. As it is mentioned before, when the slip constraint λ reaches a value of 1, it means that the wheels have lost traction and are no longer able to transmit the necessary forces to maintain the desired motion. At this point, the simulation must stop to prevent unrealistic or unsafe behavior.

The behavior of the slip constraint depends on various factors, including the surface friction, the vehicle's speed, and the steering angle. Surfaces with high friction, such as dry asphalt or concrete, typically allow for higher slip limits, and the simulation can continue for longer periods before reaching the slip constraint λ . However, on surfaces with lower friction, such as wet or icy roads, the slip constraint λ is reached much sooner, and the simulation must stop earlier to prevent unsafe conditions. The ability to accurately model the effects of slip and traction is essential for designing control systems that can maintain stability and safety during a maneuver. The Simulink model provides a powerful tool for analyzing and optimizing the behavior of the vehicle under different driving conditions and for different parameter values. By carefully adjusting the model parameters and analyzing the results, engineers can design control systems that ensure safe and efficient operation of the vehicle in a variety of driving scenarios.

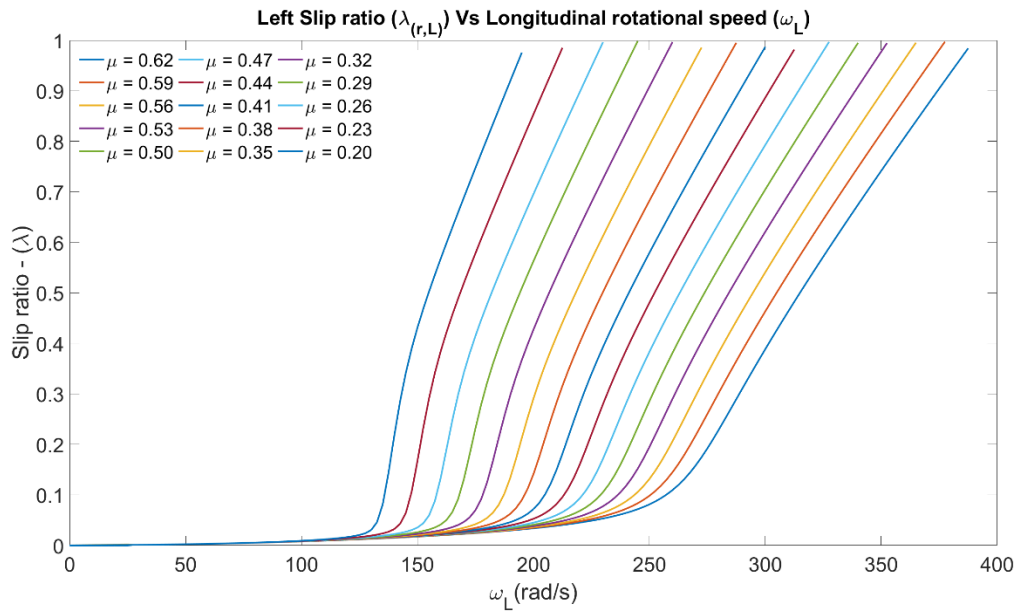


Fig.4.3.7 Left slip ratio vs w_L

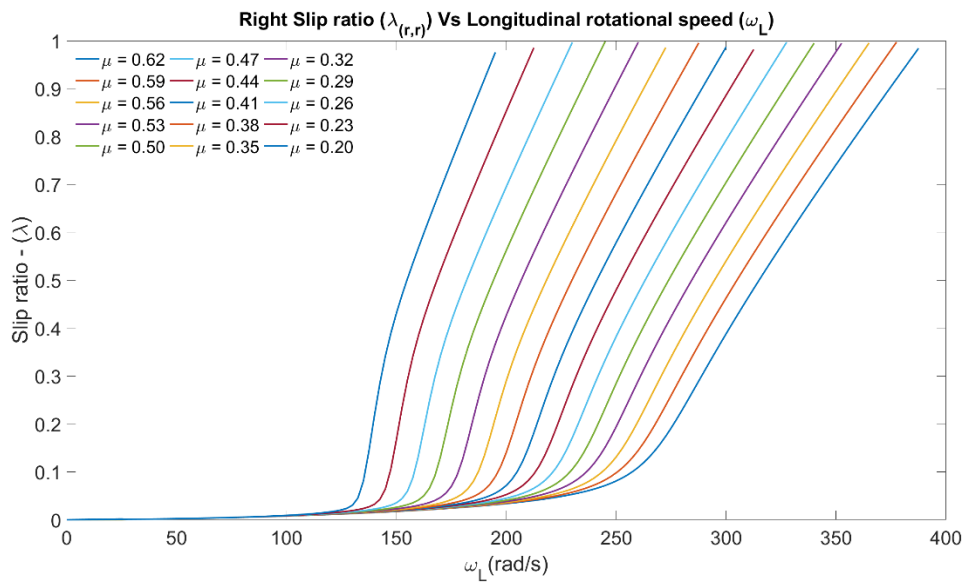


Fig.4.3.8 Right Slip Ratio vs w_L

The above images show the slip ratio – longitudinal rotational speed for both left side and right side of the car. The slip ratio λ and longitudinal speed graphs w_L for the Simulink model at different adhesion coefficient values provide a valuable insight into the performance of the vehicle under different driving conditions. The slip ratio is the ratio of the difference between the wheel speed and the vehicle speed to the wheel speed. It is a critical factor in determining the level of traction

and stability of the vehicle. As we can see from the graphs, the slip ratio reaches a value of 1 at different longitudinal speeds for different adhesion coefficient values. For example, for $\mu = 0.20$, the slip ratio reaches 1 at around 175 rad/sec, which is the lowest value among all the simulated μ values. This indicates that the vehicle loses traction and becomes unstable at lower longitudinal speeds when the adhesion coefficient is low.

On the other hand, as the adhesion coefficient value increases, the slip ratio λ values also increase, indicating that the vehicle can maintain higher longitudinal speeds without losing traction. For instance, for $\mu = 0.62$, the slip ratio reaches 1 at around 390 rad/sec, which is the highest value among all the simulated μ values. This suggests that the vehicle can maintain higher speeds and better stability on surfaces with high friction. Overall, the slip ratio and longitudinal speed graphs provide important insights into the behavior of the vehicle under different driving conditions and can help engineers design control systems that ensure stable and safe operation of the vehicle.

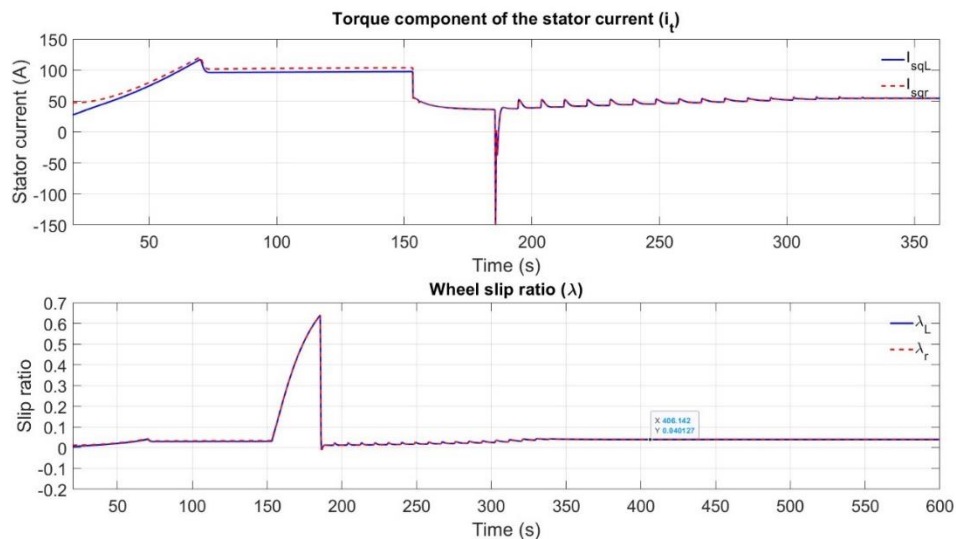


Fig.4.3.9 Wheel slip ratio and Torque

4.4) Torque Controller

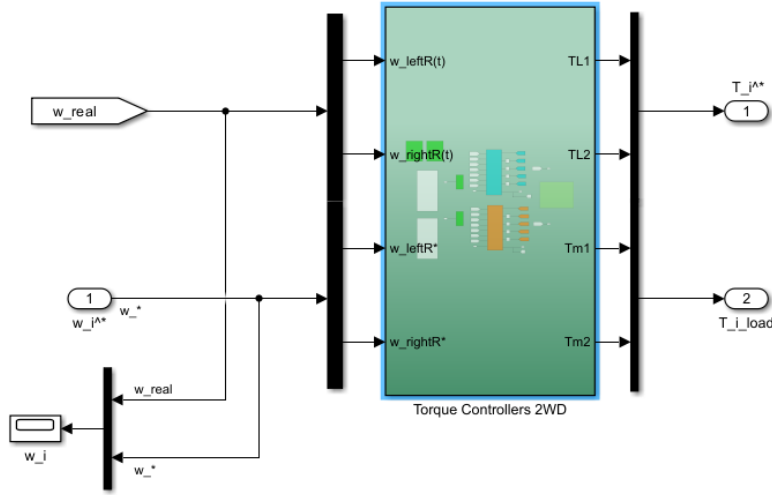


Fig.4.4.1 Torque controller

The torque controller block processes data to output the torque load, which is designed to drive the induction motors that power the left and right wheels. Induction motors are preferred over other types of motors, such as synchronous motors, due to their robustness and high reliability. They also have a high torque density and are efficient at converting electrical energy into mechanical energy, making them an ideal choice for electric car applications.

By resorting to a simplified version of the adaptive control designed, is based on a stator current control loop (k_{pi} is a positive control parameter),

$$\begin{aligned} \begin{bmatrix} u_{sa} \\ u_{sb} \end{bmatrix} &= \begin{bmatrix} \cos \varepsilon_0 & -\sin \varepsilon_0 \\ \sin \varepsilon_0 & \cos \varepsilon_0 \end{bmatrix} \begin{bmatrix} u_{sd} \\ u_{sq} \end{bmatrix} \\ u_{sd} &= -k_{pi}(i_{sd} - i_{sd}^*) + F_{fd}(\hat{T}_L^S, \hat{\alpha}_m^S, \hat{R}_s^S) \\ u_{sq} &= -k_{pi}(i_{sq} - i_{sq}^*) + F_{fq}(\hat{T}_L^S, \hat{\alpha}_m^S, \hat{R}_s^S) \\ \begin{bmatrix} i_{sd} \\ i_{sq} \end{bmatrix} &= \begin{bmatrix} \cos \varepsilon_0 & \sin \varepsilon_0 \\ -\sin \varepsilon_0 & \cos \varepsilon_0 \end{bmatrix} \begin{bmatrix} i_{sa} \\ i_{sb} \end{bmatrix}, \end{aligned}$$

The given equation defines the feedforward actions, where $F_{fd}(\cdot)$ and $F_{fq}(\cdot)$ are classical feedforward actions, and the saturated x-estimate, $\hat{X}_s$, is defined as the

output of the saturation function, $\text{sat}_{(l_1, l_2)}(\hat{X})$, which is a continuous odd function that is linear in the closed set $[l_1, l_2]$. The saturation function satisfies $\text{sat}_{(l_1, l_2)}(\hat{X}) = l_1$ for all $\hat{X} \leq l_1$ and $\text{sat}_{(l_1, l_2)}(\hat{X}) = l_2$ for all $\hat{X} \geq l_2$, where l_1 and l_2 are known lower and upper bounds on x . The reference signals (i_{sd}^* , i_{sq}^*) are responsible for rotor speed tracking and flux modulus regulation to ψ^* , respectively, along with the speed ω_0 of the (d, q) rotating frame. Overall, this equation is essential for regulating the performance and stability of the system, and it utilizes feedforward actions and saturated estimates to ensure optimal functioning of the electric motor.

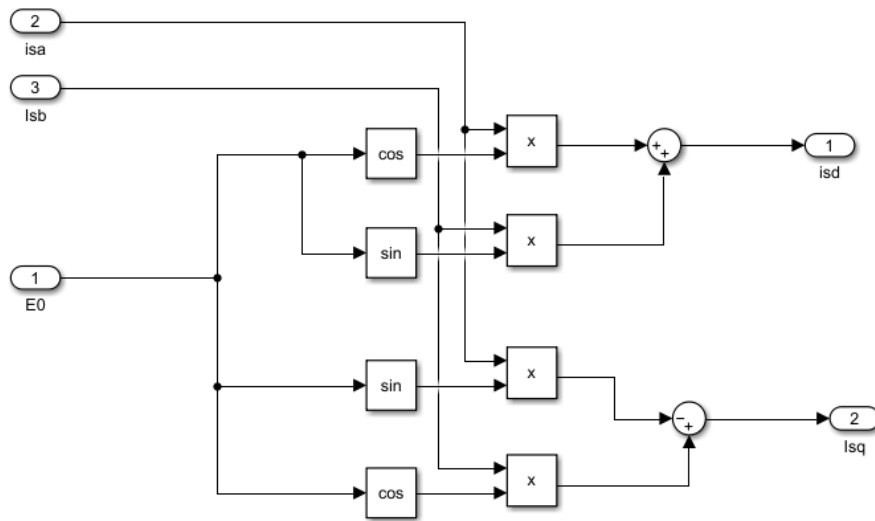


Fig.4.4.2 The Simulink model producing i_{sd} , i_{sq} , as per from the above equation.

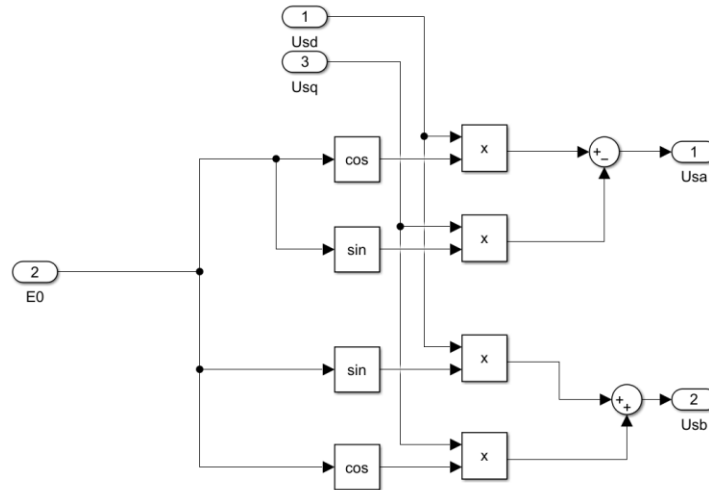


Fig.4.4.3 The Simulink model producing U_{sa} , U_{sb} , as per from the above equation.

Stator current is a critical variable in AC motor control systems, and it is typically measured in two orthogonal reference frames, known as the d-axis and the q-axis. The components of the stator current in the d- and q-axes are denoted as i_{sd} and i_{sq} , respectively. These two components are used to control the magnetic flux and torque of the motor, respectively. The stator current components can be controlled using various control strategies, such as field-oriented control (FOC) or direct torque control (DTC), to achieve optimal performance and efficiency of the motor.

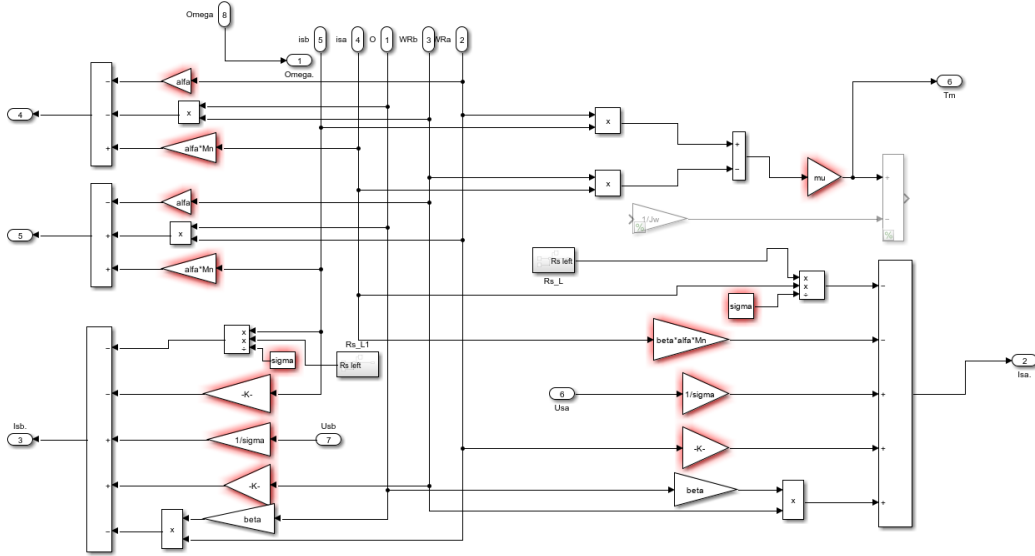


Fig.4.4.4 Equations block

The i_{sa} and i_{sb} are the stator currents, and u_{sa} and u_{sb} are the stator voltages in the fixed reference attached to the stator.

$$\frac{di_{sa}}{dt} = -\left(\frac{R_s}{\sigma} + \beta\alpha_m M\right)i_{sa} + \frac{1}{\sigma}u_{sa} + \beta\alpha_m\psi_{ra} + \beta\omega\psi_{rb}$$

$$\frac{di_{sb}}{dt} = -\left(\frac{R_s}{\sigma} + \beta\alpha_m M\right)i_{sb} + \frac{1}{\sigma}u_{sb} + \beta\alpha_m\psi_{rb} - \beta\omega\psi_{ra}$$

The rotor flux is an essential parameter in the operation of AC electric motors. It is the magnetic field produced by the rotor, which interacts with the stator field to

generate the torque required for rotation. The magnitude and orientation of the rotor flux have a significant impact on the performance of the motor, including its efficiency and power output. In an AC motor, the rotor flux is produced by the current flowing through the rotor windings, which generates a magnetic field that rotates with the rotor. The magnitude and direction of the rotor flux are influenced by various factors, such as the rotor current, the rotor position, and the magnetization characteristics of the rotor materials. The control of the rotor flux is critical for achieving optimal motor performance and efficiency, and various control techniques such as field-oriented control are used to regulate the rotor flux in real-time.

$$\begin{aligned}\frac{d\omega}{dt} &= \mu_m(\psi_{ra}i_{sb} - \psi_{rb}i_{sa}) - \frac{T_L}{J} \\ \frac{d\psi_{ra}}{dt} &= -\alpha_m\psi_{ra} - \omega\psi_{rb} + \alpha_m Mi_{sa} \\ \frac{d\psi_{rb}}{dt} &= -\alpha_m\psi_{rb} + \omega\psi_{ra} + \alpha_m Mi_{sb}\end{aligned}$$

The equation refers to a model of an electric motor, where various parameters and values are used. These include the rotor speed (ω), the rotor fluxes (ψ_{ra} , ψ_{rb}), the stator currents (i_{sa} , i_{sb}), and the stator voltages (u_{sa} , u_{sb}), all measured in a fixed reference frame attached to the stator. The model also includes several constant parameters, such as the load torque (T_L), the motor moment of inertia (J), and the rotor and stator windings resistances (R_r , R_s) and inductances (L_r , L_s), as well as the mutual inductance (M).

$$\begin{aligned}
\dot{\hat{i}}_{sa} &= -\frac{\hat{R}_s}{\sigma} i_{sa} - \hat{\alpha}_m(1 + \beta M) i_{sa} - \omega \hat{i}_{sb} + \hat{\alpha}_m \hat{z}_a \\
&\quad + \omega \hat{z}_b + \frac{u_{sa}}{\sigma} + k_i \tilde{i}_{sa} \\
\dot{\hat{i}}_{sb} &= -\frac{\hat{R}_s}{\sigma} i_{sb} - \hat{\alpha}_m(1 + \beta M) i_{sb} + \omega \hat{i}_{sa} + \hat{\alpha}_m \hat{z}_b \\
&\quad - \omega \hat{z}_a + \frac{u_{sb}}{\sigma} + k_i \tilde{i}_{sb} \\
\dot{\hat{z}}_a &= -\frac{\hat{R}_s}{\sigma} i_{sa} + \frac{u_{sa}}{\sigma} + v_a \\
\dot{\hat{z}}_b &= -\frac{\hat{R}_s}{\sigma} i_{sb} + \frac{u_{sb}}{\sigma} + v_b \\
v_a &= -k_z(\omega \tilde{i}_{sb} - \hat{\alpha}_m \tilde{i}_{sa}) \\
v_b &= k_z(\omega \tilde{i}_{sa} + \hat{\alpha}_m \tilde{i}_{sb}) \\
\dot{\hat{\alpha}}_m &= -k_\alpha [(1 + \beta M) i_{sa} - \hat{z}_a] \tilde{i}_{sa} \\
&\quad + [(1 + \beta M) i_{sb} - \hat{z}_b] \tilde{i}_{sb} \\
\dot{\hat{R}}_s &= -k_R (v_a i_{sa} + v_b i_{sb}) \\
\begin{bmatrix} \hat{\psi}_{ra} \\ \hat{\psi}_{rb} \end{bmatrix} &= -\frac{1}{\beta} \begin{bmatrix} \hat{i}_{sa} - \hat{z}_a \\ \hat{i}_{sb} - \hat{z}_b \end{bmatrix} \\
\dot{\hat{\omega}} &= \mu_m (\hat{\psi}_{ra} i_{sb} - \hat{\psi}_{rb} i_{sa}) - \frac{\hat{T}_L}{J} + k_{\omega e} e_\omega \\
\dot{\hat{T}}_L &= -k_T e_\omega
\end{aligned}$$

The estimation errors of stator current and rotor speed are essential in controlling the behavior of electric motors. These errors are determined by comparing the actual values of stator current and rotor speed with their estimated values, represented by $\hat{i}_{sa}$ and $\hat{\omega}$, respectively. The estimation errors, denoted as $\tilde{i}_{sa}$, $\tilde{i}_{sb}$, and e_ω , are then used in the control algorithm to adjust the motor's behavior.

The positive design parameters, such as k_i , k_α , $k_{\omega e}$, and k_T , are chosen to achieve desired motor behavior and performance. k_i and k_α are typically used to adjust the motor's torque output, while $k_{\omega e}$ and k_T are used to adjust the motor's speed and current output. These parameters may be adjusted based on the specific motor and control system being used.

The small positive design parameter k_R is used to ensure the stability of the control system. By choosing a small enough value for k_R , the control system can be designed to be stable and robust to disturbances.

The approach described in the equation has been studied extensively and has been shown to be effective in controlling the behavior of electric motors. Similar methods have been developed by other researchers in the field, and this approach is widely used in industrial and commercial applications.

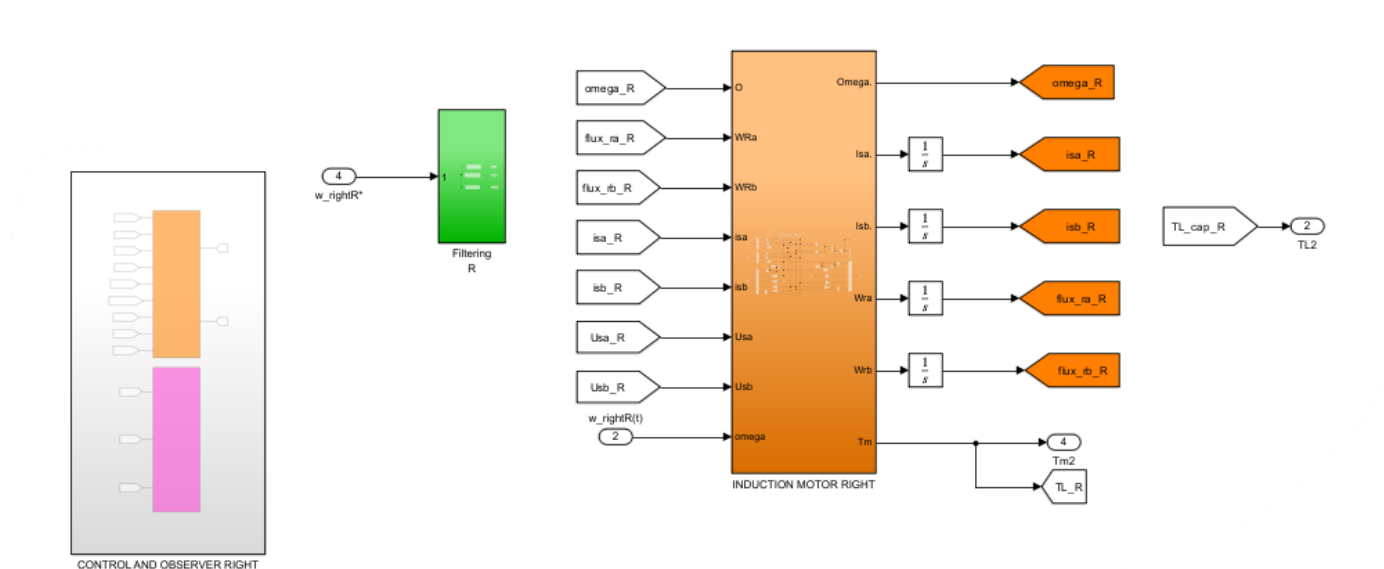


Fig.4.4.5 Induction motor

The torque generated is proportional to the current flowing through the motor windings. The motor torque current, denoted as i_t , is the current in the motor's winding that produces the torque necessary to drive the load. The torque produced by the motor is directly proportional to the motor torque current, so increasing the current will increase the motor's torque output. However, the current must be limited to prevent the motor from overheating or being damaged, and the motor torque current is typically controlled through various methods, such as current limiting or closed-loop control. The motor torque current is an important parameter to consider when designing and controlling electric motors in various applications.

4.5) End of Simulations

To generate a comprehensive visualization of the results obtained from our simulations, we need to utilize MATLAB. To achieve this, we have to undertake several steps. Firstly, after executing each simulation, it is necessary to store the generated data in a file. This is a crucial step as it ensures that all the data points obtained from the simulations are saved for future reference.

Next, a MATLAB script needs to be written to combine all the individual graphs into a single master graph. This process is important as it allows us to compare and track changes over time across all simulations. The script will enable us to automate the process of merging the graphs, making it faster and easier to analyze the results.

The merging of the graphs allows us to identify any significant patterns, trends, or anomalies that may exist in the data. These insights will help us to make more informed decisions and draw accurate conclusions about the behavior of our system. Finally, by running the new script, all of the merged graphs will be combined to create an overall view that shows how values change throughout the experiment. This will enable us to gain a more comprehensive understanding of the performance of our system and to identify any areas that require further improvement. The use of MATLAB is essential for creating a comprehensive visualization of the results obtained from simulations. By following the steps mentioned above, we can ensure that all the data is saved and analyzed in a systematic and automated manner, leading to more accurate and insightful results.

Some of the graphs are below,

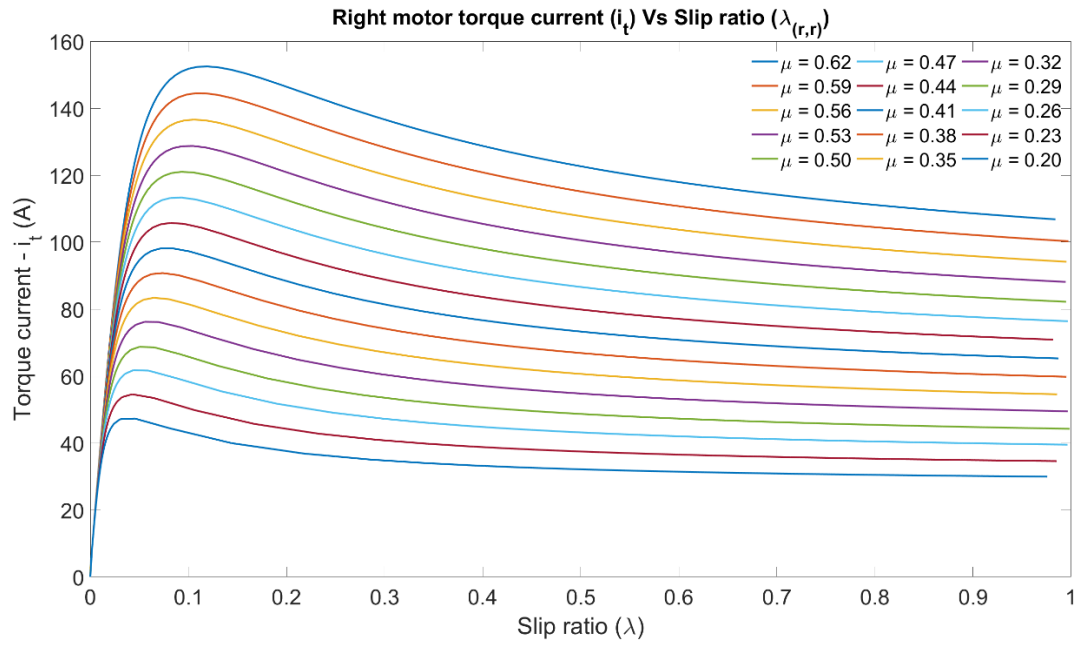


Fig.4.5.1 Right motor torque vs slip ratio

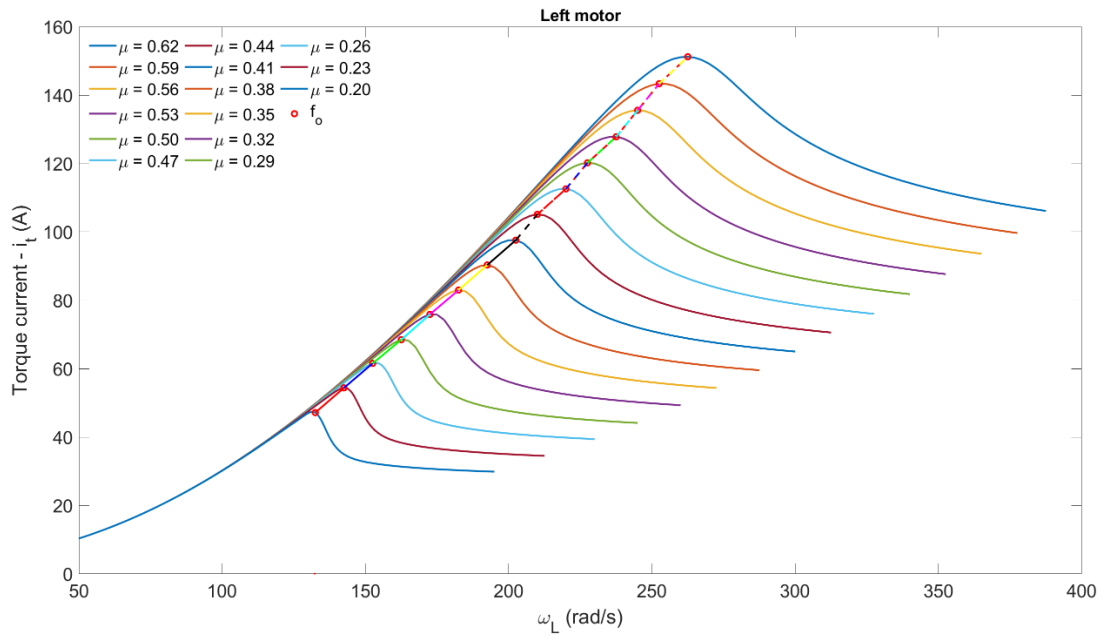


Fig.4.5.2 Left motor curve

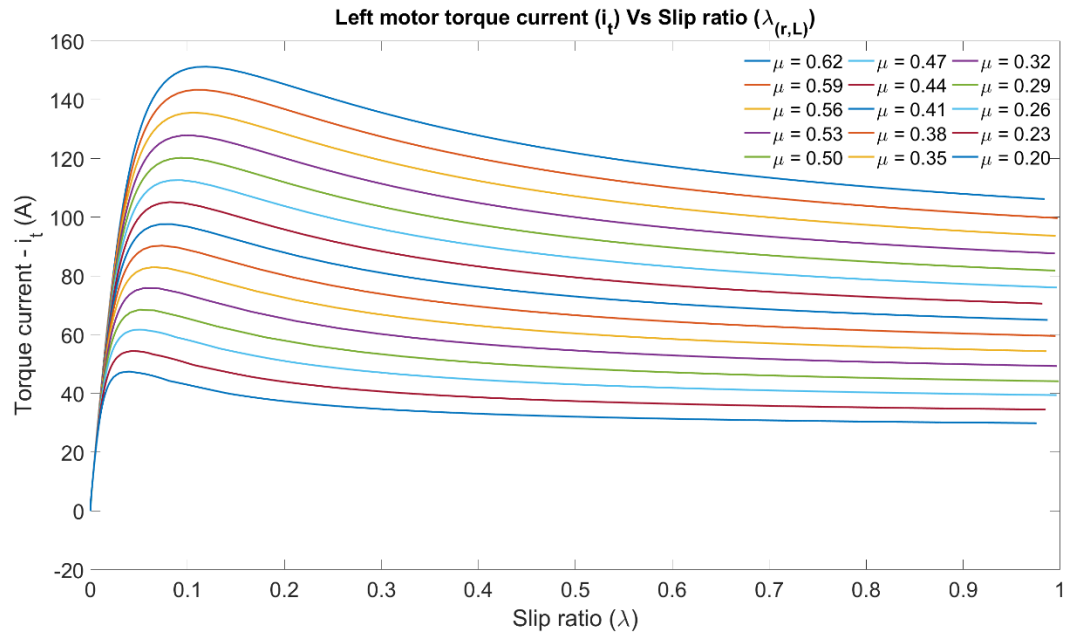


Fig.4.5.3 Left motor torque vs slip ratio

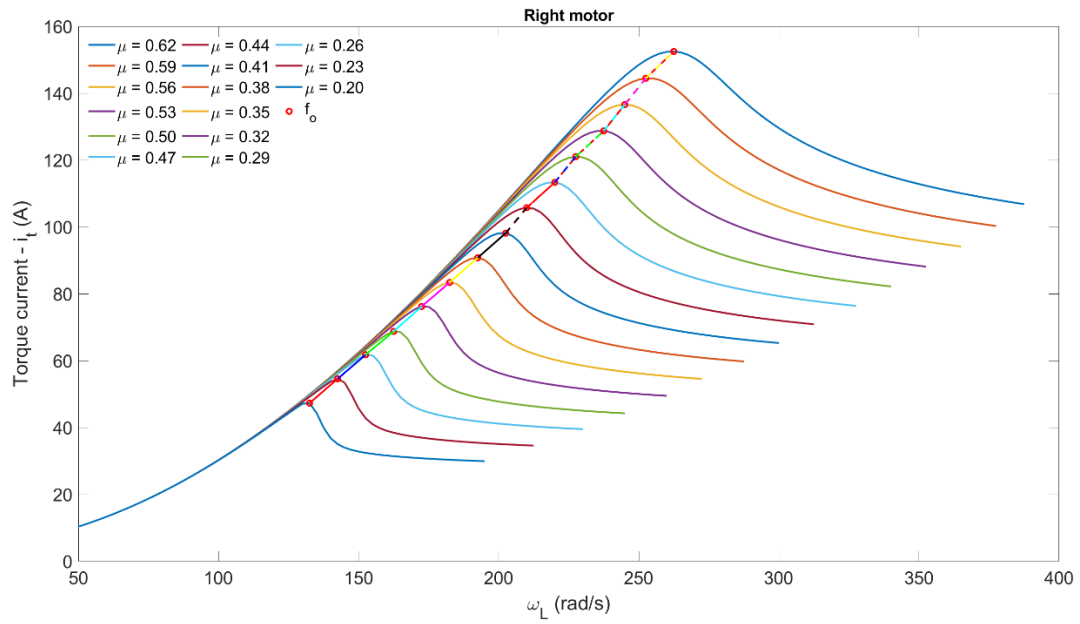


Fig.4.5.4 Right motor

The difference in torque between the right and left motors during a slow left turn is a result of the mechanics of the vehicle. When a car is turning left, the weight of the car shifts towards the right side, causing the right tires to grip the road surface more firmly. This increased traction results in greater torque being applied to the right side of the vehicle, which is reflected in the torque curve of the right motor.

Moreover, the difference in torque between the right and left motors can also be attributed to the steering mechanism of the car. During a left turn, the steering system applies a greater steering angle to the left wheel, which in turn reduces its torque output. This reduction in torque output is reflected in the torque curve of the left motor.

By analyzing the Torque current, and Longitudinal Speed graphs, we can gain valuable insights into the behavior of the vehicle during a slow left turn. The slight difference in torque between the right and left motors may seem insignificant, but it can have a significant impact on the overall performance of the vehicle. Therefore, it is essential to monitor these parameters carefully to ensure optimal vehicle performance and safety.

4.6) The Results

The theorem states, If we have a finite number of typical $\chi_{\mu,c}(\cdot)$ -curves, we can still use Proposition 1 even if we relax constraint 2) and construct $f_O(\cdot)$ by linearly interpolating the points $P_{\max,\mu}$ that correspond to the unique maximum of $\chi_{\mu,c}(\cdot)$ at different μ . By using a finite number of T-iteration steps, we can ensure that we approach $P_{\max,\mu}$ from the left in a finite amount of time. This approach provides a safe operating condition that is close to the maximum of the torque-slip characteristics, without requiring knowledge of the tire-road adhesion coefficient.

Based on the previous subsection, it has been found that the procedure used is structurally conservative. This has been demonstrated through simulations conducted in the CarSim environment, which take into account complex vehicle dynamics that were not considered during the design stage. The CarSim vehicle

model includes detailed nonlinear tire models that consider both lateral and longitudinal forces, using combined slip theory. Additionally, the model accounts for major kinematics and compliance effects of the suspensions (using nonlinear spring models) and steering systems. The simulations were conducted using a specific vehicle model, which is a GT rear-wheel drive car with independent front and rear suspension systems. The car's dimensions include a height of 984 mm, length of 1628 mm, and width of 1240 mm. The car weighs 486 kg and has a wheelbase of 0.838 m (l_r) and 0.79 m (l_f). The front and rear wheels are 175/70 R13 with a radius of 0.225 m (R_w). The transmission gear ratio is fixed, and the vehicle closely follows a prescribed path of 12000 m in the x-direction and 1400 m in the y-direction.

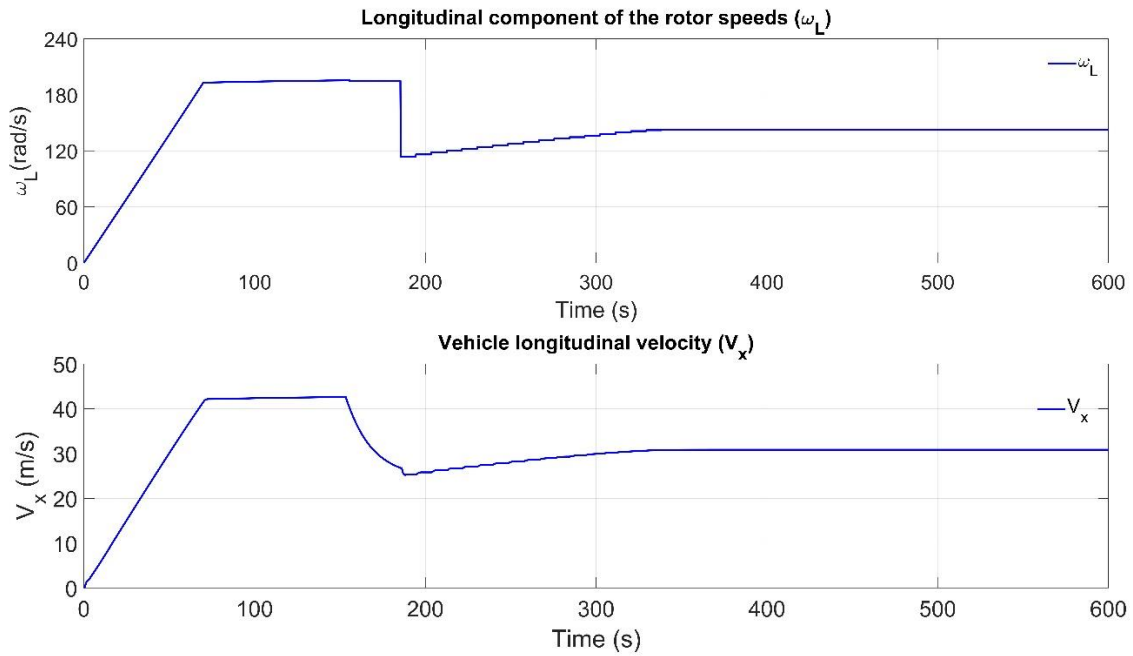


Fig.4.6.1 Longitudinal components of the rotor speed and Vehicle longitudinal velocity

The filter time constants used in this study are $a_\omega = 9$ s and $b_\omega = 8.99$ s. The subsequent filter gains $\gamma_1, \gamma_2, \gamma_3$ have been selected such that all the eigenvalues of the matrix characterizing the π i-dynamics are equal to -10. These filters are applied at the stage where the rotor speed reference is defined. The vehicle is powered by two rear 37 kW induction motors (operating at 50 Hz, 400/230 V, 64/111 A,

generating 24.17 Nm of torque at 2960 rpm), whose parameters are provided in the literature. The motors are controlled using field-oriented control, where the rotor flux modulus reference is set to 1.4 Wb. Initially, the vehicle is controlled using a control algorithm with no feedforward terms and PI gains of (25, 15) and (20, 10), respectively.

The control design proposed in this study utilizes a clever application of Pacejka-like curves in the (ω_L, i_t) -plane. Here, " i_t " represents the motor torque current, which is the current component of each electrical motor that controls the speed of the vehicle. The set " j " includes the pairs (r, l) and (r, r) . To describe the scenario underlying the proposed design, the study introduces an assumption that involves an uncertain parameter vector μ , which belongs to the set $M \subset \mathbb{R}^3$. This vector takes the same value for each driving wheel and includes a description of external conditions. This assumption forms the basis for the design of the control algorithm that can adapt to uncertain external conditions and ensure stability and good performance of the vehicle.

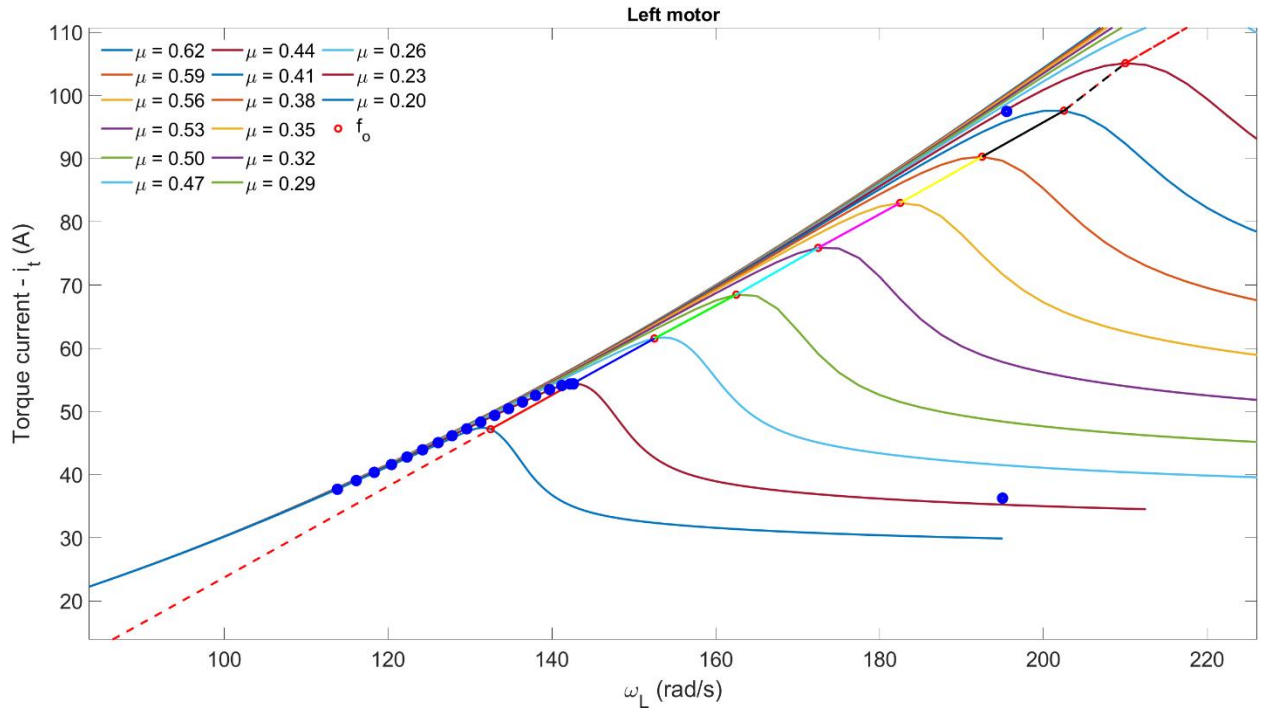


Fig.4.6.2 Left motor curve

The algorithm is designed to tightly follow a ramp/constant-wise vehicle speed reference, which reaches a constant profile at a speed of $v_x^* = 43$ m/s (154.8 km/h) at approximately $t = 70$ s. At this operation point, the rotor speed is $\omega_L = 198.176$ rad/s, and the slip ratio for both wheels is about 0.037. The vehicle is assumed to be moving on the road with a coefficient of friction of $\mu[1] = 0.60$. Then a first abrupt change in the road conditions appears at the third km, with the tire-road adhesion coefficient $\mu[1]$ reducing from $\mu[1] = 0.60$ to $\mu[1] = 0.23$.

The approach used in the study relies on the contraction theory and involves the use of Pacejka-like curves in the (ω_L, i_t) -plane, where it represents the motor torque current. This current component of each electrical motor is responsible for speed control. The study assumes an uncertain parameter vector $\mu \in M \subset \mathbb{R}^3$ that includes the description of external conditions and takes the same value for each driving wheel. Contraction mapping algorithm is used to decrease the speed in correct way.

When the left tire longitudinal slip reaches the limit value of 0.1, the ω_L value is maintained at the corresponding pre-variation value for 30 seconds. This operation point is beyond the maximum of the 0.23-curve, which is a curve that characterizes the maximum value of longitudinal force that can be generated by the tire.

The contraction theorem-based approach proposed in the study guarantees the convergence to the intersection point $P_{\mu,c} | \mu[1]=0.23$ between the function $g_{\mu,c} | \mu[1]=0.23(\cdot)$ and the function $f_{sl}(\omega_L)$. This intersection point is located at $\omega_L = 153.66$ rad/s, $v_x = 32.94$ m/s (118.59 km/h), with a slip ratio approximately equal to 0.05 for both wheels.

The resulting profiles of vehicle speed (v_x) and ω_L are presented in the figures reported in the study. The approach is shown to be effective in maintaining the vehicle's speed within the desired range while ensuring the safety of the vehicle by preventing the longitudinal slip from exceeding the maximum value. The study presents a novel and effective approach for controlling vehicle speed in the presence of uncertain external conditions, and it demonstrates the usefulness of the contraction theory-based approach for vehicle control problems.

5) Conclusion

The presented solution for electric vehicles with two in-wheel motors and the need to perform a bend manoeuvre with a small steering angle is innovative. The solution is based on Theorem and a finite collection of typical $\chi_{\mu,c}(\cdot)$ -curves. Proposition 1 still holds for such μ when constraint is relaxed, and $f_o(\cdot)$ is constructed by linearly interpolating the points $P_{\max,\mu}$ that constitute the unique maxima of $\chi_{\mu,c}(\cdot)$ at different μ . Using a finite number of T-iteration steps ensures a finite-time approach to $P_{\max,\mu}$ from the left, guaranteeing a safe operating condition that is close to the maximum of the torque-slip characteristics without requiring knowledge of the tire-road adhesion coefficient. In the simulation described above, the vehicle was initially moving at a constant value of μ , without any external interference. However, when the value of μ suddenly changed, the vehicle's speed was drastically affected. For instance, when μ changed from 0.60 to 0.23, the speed of the vehicle decreased steeply from 0.60 to 0.23 on the graph. This change in the value of μ could be due to external factors such as a change in the road surface or weather conditions. To address this sudden change in speed, slip constraints were employed to gradually increase the vehicle speed and longitudinal speed while stabilizing the vehicle. The approach taken in the simulation was effective in ensuring that the vehicle remained safe and stable, without causing any further problems.

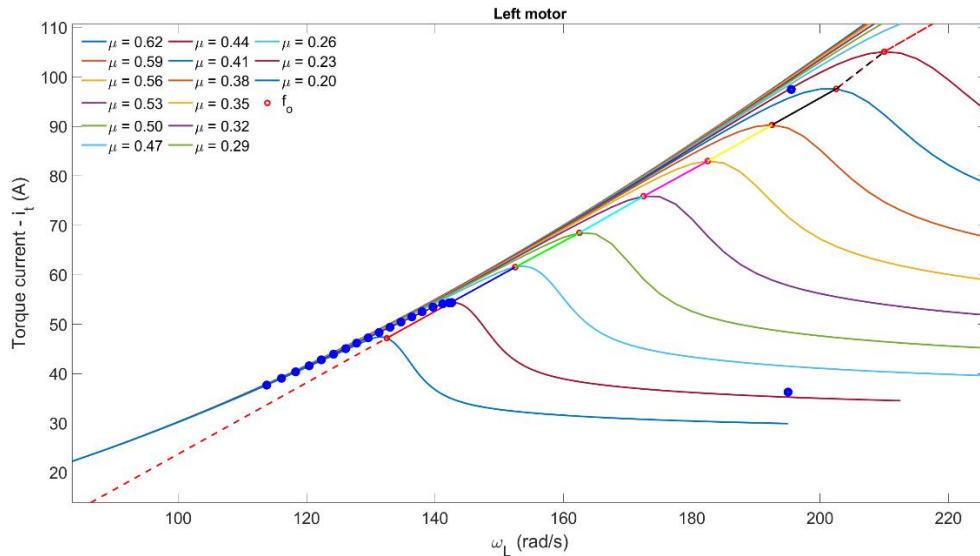


Fig.5.1 Pacejka Like curve-Left motor

The ability to adapt to changes in external conditions and maintain stability is crucial for electric vehicles. This simulation highlights the importance of incorporating advanced control algorithms and slip constraints to ensure the safety and stability of electric vehicles, particularly in situations where external conditions may change suddenly and unpredictably. We can observe an abrupt change in the road conditions as the value of $\mu[1]$ drops from $\mu[1] = 0.60$ to $\mu[1] = 0.23$. To handle this change, a new control approach is used where iteration points on the $(\omega_L, i, (r, l))$ -plane are calculated for both values of $\mu[1]$. When the longitudinal slip of the left tire reaches the limit value of 0.1, the ω_L is initially kept constant for 30 seconds at its pre-variation value of 198.176 rad/s, and then modified based on the motor speed reference generator. As per the analysis based on the aforementioned theorem, the longitudinal speed of the vehicle iteratively decreases from 43 m/s (154.8 km/h) to 25 m/s, which corresponds to the steady-state operation point at the maximum of the 0.23-curve that comes just before the intersection point $P_{\mu,c} | \mu[1]=0.23$. Due to this change in longitudinal speed, the tire longitudinal slip also decreases to the corresponding value on the characteristic. At around 262 seconds (eighth km), there is a sudden increase in the tire-road adhesion coefficient $\mu[1]$ back to $\mu[1] = 0.23$.

Related Figures

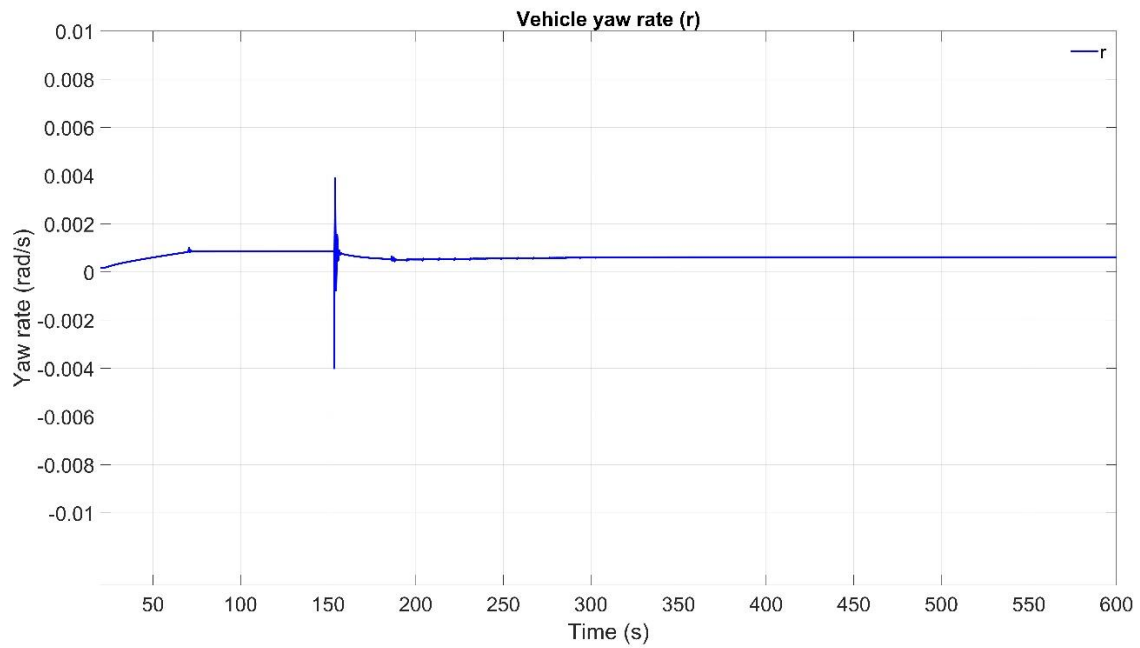


Fig.6.1 Vehicle yaw rate

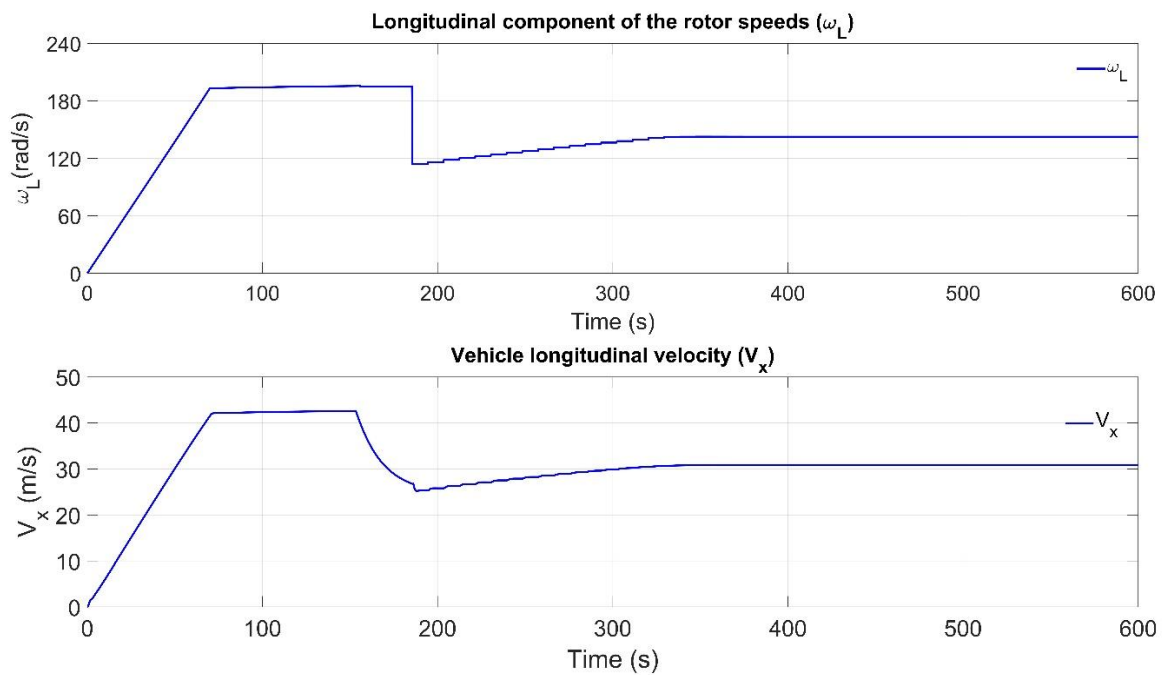


Fig.6.2 w_L and v_x vs time

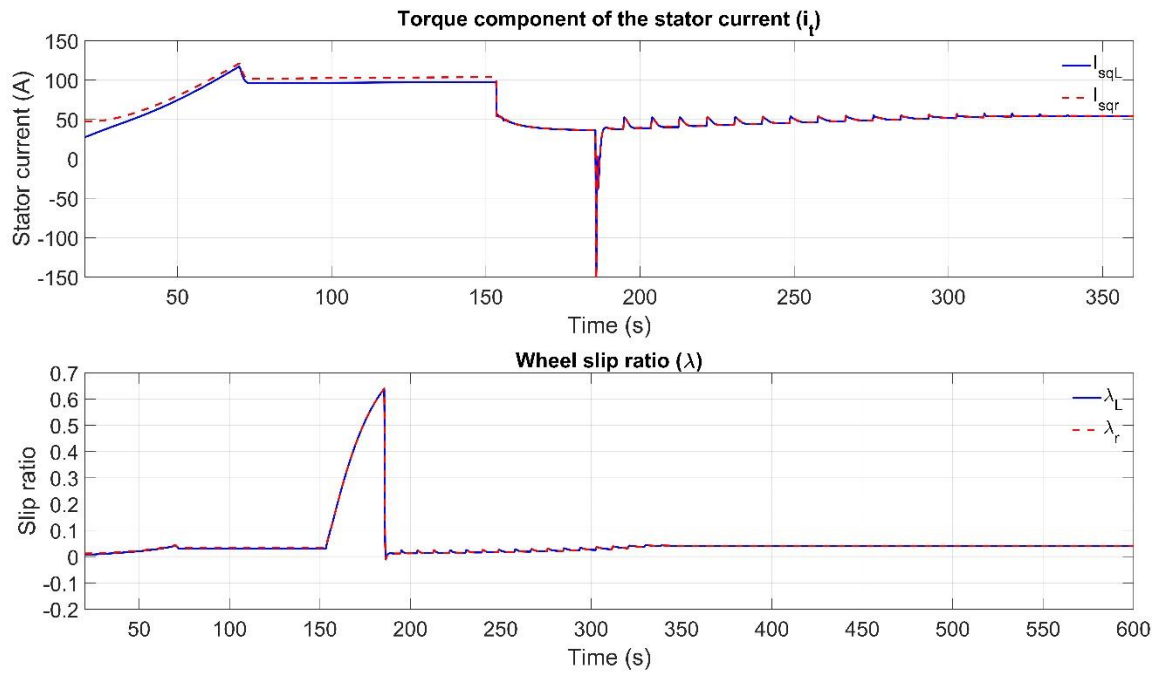


Fig.6.3 Torque component of stator current and Wheel slip ratio

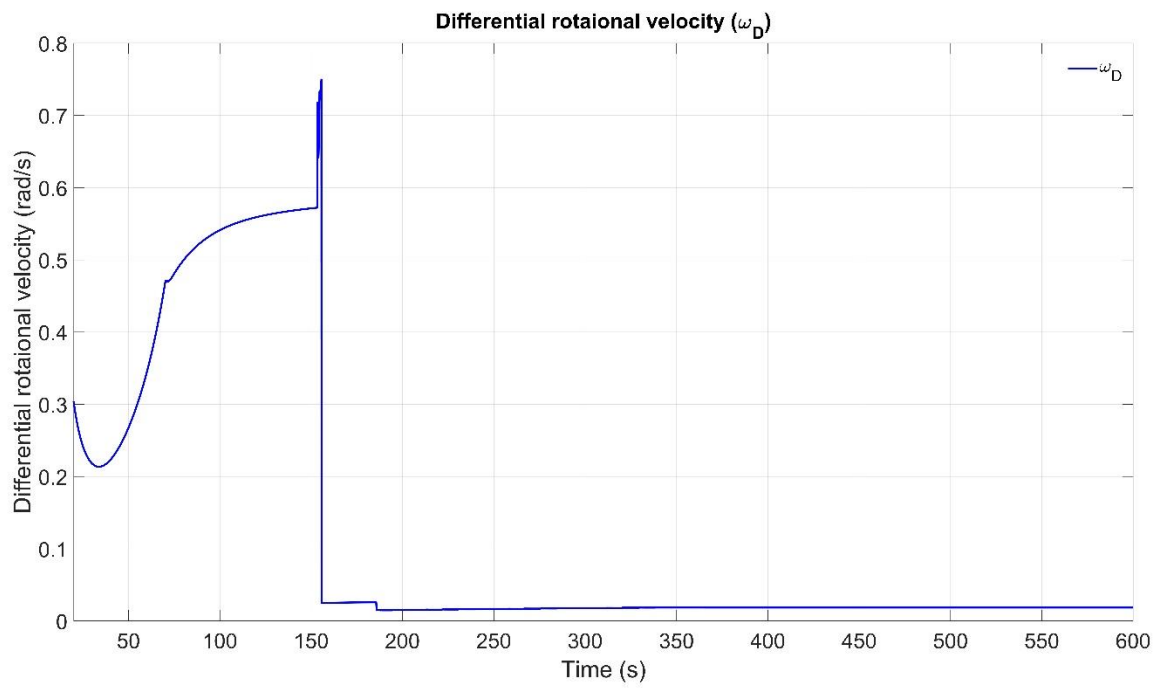


Fig.6.4 Differential rotational velocity

References

- [1] C. M. Verrelli and M. El Arayshi, "Pacejka-Like Curve-Based Speed Reference Generators for Electric Vehicles Powered by in-Wheel Motors" revised 22 November 2022, Date of publication 27 December 2022.
- [2] R. Marino, P. Tomei, and C. M. Verrelli, *Induction Motor Control Design*. London, U.K.: Springer, 2010.
- [3] P. Tomei and C. M. Verrelli, "Observer-based speed tracking control for sensorless permanent magnet synchronous motors with unknown load torque," *IEEE Trans. Autom. Control*, vol. 56, no. 6, pp. 1484–1488, Jun. 2011.
- [4] Y. Hori, "Future vehicle driven by electricity and control—Research on four-wheel-motored 'UOT Electric March II,'" *IEEE Trans. Ind. Electron.*, vol. 51, no. 5, pp. 954–962, Oct. 2004. [4] H. Alipour, M. Sabahi, M. B. B. Sharifian, "Lateral stabilization of a four wheel independent drive electric vehicle on slippery roads," *Mechatronics*, vol. 30, pp. 275–285, Sep. 2015.
- [5] G. Zhang, H. Zhang, X. Huang, J. Wang, H. Yu, and R. Graaf, "Active fault-tolerant control for electric vehicles with independently driven rear in-wheel motors against certain actuator faults," *IEEE Trans. Control Syst. Technol.*, vol. 24, no. 5, pp. 1557–1572, Sep. 2016.
- [6] Y.-P. Yang and X.-Y. Xing, "Design of electric differential system for an electric vehicle with dual wheel motors," in *Proc. 47th IEEE Conf. Decis. Control*, 2008, pp. 4414–4419.
- [7] C. Pomponi, S. Scalzi, L. Pasquale, C. M. Verrelli, and R. Marino, "Automatic motor speed reference generators for cruise and lateral control of electric vehicles with in-wheel motors," *Control Eng. Pract.*, vol. 79, pp. 126–143, Oct. 2018.
- [8] M. Reichhartinger and M. Horn, "Robust cascaded automatic cruise control of electric vehicles," *Int. J. Robust Nonlinear Control*, vol. 26, no. 4, pp. 662–680, 2016, doi: 10.1002/rnc.3330.
- [9] Y. Luo, T. Chen, S. Zhang, and K. Li, "Intelligent hybrid electric vehicle ACC with coordinated control of tracking ability, fuel economy, and ride comfort," in *IEEE Trans. Intell. Transp. Syst.*, vol. 16, no. 4, pp. 2303–2308, Aug. 2015.
- [10] J.E. Naranjo, C. González, R. García, and T. de Pedro, "ACC+Stop&Go maneuvers with throttle and brake fuzzy control," *IEEE Trans. Intell. Transp. Syst.*, vol. 7, no. 2, pp. 213–225, Jun. 2006.

- [11] Y. Chen and J. Wang, "Adaptive vehicle speed control with input injections for longitudinal motion independent road frictional condition estimation," *IEEE Trans. Veh. Technol.*, vol. 60, no. 3, pp. 839–848, Mar. 2011.
- [12] J. E. Naranjo, C. González, J. Reviejo, R. García, and T. de Pedro, "Adaptive fuzzy control for inter-vehicle gap keeping," *IEEE Trans. Intell. Transp. Syst.*, vol. 4, no. 3, pp. 132–142, Sep. 2003.
- [13] V. L. Bageshwar, W. L. Garrard, and R. Rajamani, "Model predictive control of transitional maneuvers for adaptive cruise control vehicles," *IEEE Trans. Veh. Technol.*, vol. 53, no. 5, pp. 1573–1585, Sep. 2004.
- [14] J. Wang, X. Xu, D. Liu, Z. Sun, and Q. Chen, "Self-learning cruise control using kernel-based least squares policy iteration," *IEEE Trans. Control Syst. Technol.*, vol. 22, no. 3, pp. 1078–1087, May 2014.
- [15] S. H. Hosseinnia, I. Tejado, V. Milanés, J. Villagrà, and B. M. Vinagre, "Experimental application of hybrid fractional-order adaptive cruise control at low speed," *IEEE Trans. Control Syst. Technol.*, vol. 22, no. 6, pp. 2329–2336, Nov. 2014.
- [16] C. Desjardins and B. Chaib-draa, "Cooperative adaptive cruise control: A reinforcement learning approach," *IEEE Trans. Intell. Transp. Syst.*, vol. 12, no. 4, pp. 1248–1260, Dec. 2011.
- [17] R. Rajamani and C. Zhu, "Semi-autonomous adaptive cruise control systems," *IEEE Trans. Veh. Technol.*, vol. 51, no. 5, pp. 1186–1192, Sep. 2002.
- [18] D. Corona and B. De Schutter, "Adaptive cruise control for a SMART car: A comparison benchmark for MPC-PWA control methods," *IEEE Trans. Control Syst. Technol.*, vol. 16, no. 2, pp. 365–372, Mar. 2008.
- [19] B. Subudhi and S. S. Ge, "Sliding-mode-observer-based adaptive slip ratio control for electric and hybrid vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 13, no. 4, pp. 1617–1626, Dec. 2012.
- [20] M. Yamada, K. Ueda, I. Horiba, and N. Sugie, "Discrimination of the road condition toward understanding of vehicle driving environments," *IEEE Trans. Intell. Transp. Syst.*, vol. 2, no. 1, pp. 26–31, Mar. 2001.
- [21] R. Marino, S. Scalzi, P. Tomei, and C. M. Verrelli, "Fault-tolerant cruise control of electric vehicles with induction motors," *Control Eng. Pract.*, vol. 21, pp. 860–869, Jun. 2013.
- [22] R. Marino, L. Pasquale, S. Scalzi, and C. M. Verrelli, "Automatic rotor speed reference generator for electric vehicles under slip constraints," *IEEE Trans. Intell. Transp. Syst.*, vol. 16, no. 6, pp. 3473–3478, Dec. 2015.
- [23] E. Coelingh and S. Solyom, "All aboard the robotic road train," *IEEE Spectr.*, vol. 49, no. 11, pp. 34–39, Nov. 2012.

- [24] G. Amato and R. Marino, "Reconfigurable slip vectoring control in four in-wheel drive electric vehicles," *Actuators*, vol. 10, p. 157, Jul. 2021.
- [25] M. El Arayshi, G. Amato, and F. Pischiutta, "A new monitoring strategy and automatic speed reduction for PI slip control in autonomous electric vehicles with in-wheel motors: Experimental results," in *Proc. Int. Symp. Power Electron. Electr. Drives, Autom. Motion (SPEEDAM)*, 2022, pp. 820–825.
- [26] H. Du, N. Zhang, and F. Naghdy, "Velocity-dependent robust control for improving vehicle lateral dynamics," *Transp. Res. C*, vol. 19, pp. 454–468, Jun. 2011.
- [27] F. Tahami, R. Kazemi, and S. Farhanghi, "A novel driver assist stability system for all-wheel-drive electric vehicles," *IEEE Trans. Veh. Technol.*, vol. 52, no. 3, pp. 683–692, May 2003.
- [28] A. El Magri, F. Giri, G. Besancon, A. El Fadili, L. Dugardm, and ´ F. Z. Chaoui, "Sensorless adaptive output feedback control of wind energy systems with PMS generators," *Control Eng. Pract.*, vol. 21, pp. 530–543, Apr. 2013.
- [29] E. Kreyszig, *Introductory Functional Analysis With Applications*. New York, NY, USA: Wiley, 1978.
- [30] M. Reed and B. Simon, *Functional Analysis (Methods of Modern Mathematical Physics)*, vol. 1. London, U.K.: Acad. Press, 1980.
- [31] H. K. Khalil. *Nonlinear Systems*. Upper Saddle River, NJ, USA: Prentice Hall, 2002.
- [32] M. Tanelli, A. Astolfi, and S. M. Savaresi, "Robust nonlinear output feedback control for brake by wire control systems," *Automatica*, vol. 44, pp. 1078–1087, Apr. 2008.
- [33] M. Aguado-Rojas, W. Pasillas-Lépine, and A. Loría, "Extendedbraking-stiffness estimation under varying road-adherence conditions," *IEEE Trans. Control Syst. Technol.*, vol. 28, no. 5, pp. 1964–1971, Sep. 2020.